

genetic_algorithm

August 28, 2024

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[1]: import random
import numpy as np
import pandas as pd
import joblib

[2]: #my local path
df = pd.read_excel("/home/houssam/internship project/Ammonia production final_
↳data.xlsx")
#creat data
data = {
    "NG FEED - Comp Mass Flow (KJ/h)": df.loc[2:, 'Unnamed: 3'],
    "Ratio (S/C)": df.loc[2:, 'Unnamed: 4'],
    "Mixed Steam- Comp Mass Flow (KJ/h)": df.loc[2:, 'Unnamed: 5'],
    "Reformer inlet temperature (°C)": df.loc[2:, 'Unnamed: 6'],
    "Reformer inlet pressure (kPa)": df.loc[2:, 'Unnamed: 7'],
    "Reformer outlet temperature (°C)": df.loc[2:, 'Unnamed: 8'],
    "Reformer outlet pressure (kPa)": df.loc[2:, 'Unnamed: 9'],
    "Reformer Energy consumption (KJ/h)": df.loc[2:, 'Unnamed: 10'],
    "HTS - Inlet temperature (°C)": df.loc[2:, 'Unnamed: 11'],
    "HTS - Inlet pressure (kPa)": df.loc[2:, 'Unnamed: 12'],
    "HTS - Outlet temperature (°C)": df.loc[2:, 'Unnamed: 13'],
    "HTS - Outlet pressure (kPa)": df.loc[2:, 'Unnamed: 14'],
    "E-HTS - Energy consumption (KJ/h)": df.loc[2:, 'Unnamed: 15'],
    "LTS - Inlet temperature (°C)": df.loc[2:, 'Unnamed: 16'],
    "LTS - Inlet pressure (kPa)": df.loc[2:, 'Unnamed: 17'],
    "LTS - Outlet temperature (°C)": df.loc[2:, 'Unnamed: 18'],
    "LTS - Outlet pressure (kPa)": df.loc[2:, 'Unnamed: 19'],
    "E-LTS - Energy consumption (KJ/h)": df.loc[2:, 'Unnamed: 20'],
    "Condensor inlet temperature (°C)": df.loc[2:, 'Unnamed: 21'],
    "Condensor inlet pressure (kPa)": df.loc[2:, 'Unnamed: 22'],
    "Condensor outlet temperature (°C)": df.loc[2:, 'Unnamed: 23'],
    "Condensor outlet pressure (kPa)": df.loc[2:, 'Unnamed: 24'],
    "E-Condensor- Energy consumption (KJ/h)": df.loc[2:, 'Unnamed: 25'],
    "CO2 emissions-Comp Mass Flow (KJ/h)": df.loc[2:, 'Unnamed: 26'],
    "CO2 removal- Energy consumption (KJ/h)": df.loc[2:, 'Unnamed: 27'],
    "Methanator inlet temperature (°C)": df.loc[2:, 'Unnamed: 28'],
    "Methanator-Outlet temperature (°C)": df.loc[2:, 'Unnamed: 29'],
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"Methanator-Outlet pressure (kPa)": df.loc[2:, 'Unnamed: 30'],
"Methanator-Energy consumption (KJ/h)": df.loc[2:, 'Unnamed: 31'],
"Hydrogen inlet temperature (°C)": df.loc[2:, 'Unnamed: 32'],
"Hydrogen inlet pressure (kPa)": df.loc[2:, 'Unnamed: 33'],
"Nitrogen inlet temperature (°C)": df.loc[2:, 'Unnamed: 34'],
"Nitrogen inlet pressure (kPa)": df.loc[2:, 'Unnamed: 35'],
"S-1 stream temperature (°C)": df.loc[2:, 'Unnamed: 36'],
"S-1 stream pressure (kPa)": df.loc[2:, 'Unnamed: 37'],
"K-100 - Delta P (kPa)": df.loc[2:, 'Unnamed: 38'],
"S-2 stream temperature (°C)": df.loc[2:, 'Unnamed: 39'],
"S-2 stream pressure (kPa)": df.loc[2:, 'Unnamed: 40'],
"S-3 stream temperature (°C)": df.loc[2:, 'Unnamed: 41'],
"S-3 stream pressure (kPa)": df.loc[2:, 'Unnamed: 42'],
"K-102 - Delta P (kPa)": df.loc[2:, 'Unnamed: 43'],
"S-4 stream pressure (kPa)": df.loc[2:, 'Unnamed: 44'],
"Synthesis gas inlet temperature (°C)": df.loc[2:, 'Unnamed: 45'],
"Synthesis gas inlet pressure (kPa)": df.loc[2:, 'Unnamed: 46'],
"Synthesis gas outlet temperature (°C)": df.loc[2:, 'Unnamed: 47'],
"Synthesis gas outlet pressure (kPa)": df.loc[2:, 'Unnamed: 48'],
"H.B reactor Energy consumption (KJ/h)": df.loc[2:, 'Unnamed: 49'],
"Separator inlet temperature (°C)": df.loc[2:, 'Unnamed: 50'],
"Separator inlet pressure (kPa)": df.loc[2:, 'Unnamed: 51'],
"Separator outlet pressure (kPa)": df.loc[2:, 'Unnamed: 52'],
"E-separator- Energy consumption (KJ/h)": df.loc[2:, 'Unnamed: 53'],
"Q-100 - Energy consumption (KJ/h)": df.loc[2:, 'Unnamed: 54'],
"Q-102 - Energy consumption (KJ/h)": df.loc[2:, 'Unnamed: 55'],
"Q-103 - Energy consumption (KJ/h)": df.loc[2:, 'Unnamed: 56'],
"Q-104 - Energy consumption (KJ/h)": df.loc[2:, 'Unnamed: 57'],
"Q-105- Energy consumption (KJ/h)": df.loc[2:, 'Unnamed: 58'],
"E-1- Energy consumption (KJ/h)": df.loc[2:, 'Unnamed: 59'],
"E-2- Energy consumption (KJ/h)": df.loc[2:, 'Unnamed: 60'],
"E-3- Energy consumption (KJ/h)": df.loc[2:, 'Unnamed: 61'],
"E-4- Energy consumption (KJ/h)": df.loc[2:, 'Unnamed: 62'],
"Ammonia - Comp Mass Fraction": df.loc[2:, 'Unnamed: 63'],
"Ammonia - Comp Mass Flow (kg/h)": df.loc[2:, 'Unnamed: 64']
}
df = pd.DataFrame(data)
df = df.apply(pd.to_numeric, errors='coerce')

energy_columns = [
    "Reformer Energy consumption (KJ/h)",
    "E-HTS - Energy consumption (KJ/h)",
    "E-LTS - Energy consumption (KJ/h)",
    "E-Condensor- Energy consumption (KJ/h)",
    "CO2 removal- Energy consumption (KJ/h)",
    "Methanator-Energy consumption (KJ/h)",
    "H.B reactor Energy consumption (KJ/h)",

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"E-separator- Energy consumption (KJ/h)",
"Q-100 - Energy consumption (KJ/h)",
"Q-102 - Energy consumption (KJ/h)",
"Q-103 - Energy consumption (KJ/h)",
"Q-104 - Energy consumption (KJ/h)",
"Q-105- Energy consumption (KJ/h)",
"E-1- Energy consumption (KJ/h)",
"E-2- Energy consumption (KJ/h)",
"E-3- Energy consumption (KJ/h)",
"E-4- Energy consumption (KJ/h)"
]

df['total_energy'] = df[energy_columns].sum(axis=1)

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[3]: import warnings
warnings.filterwarnings(
    "ignore",
    category=UserWarning,
    message="X does not have valid feature names"
)

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[ ]: #genetic algo with fixed ranges

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[4]: #define the population size and number of generations
POPULATION_SIZE = 100
NUM_GENERATIONS = 500
MUTATION_RATE = 0.5
ELITISM_SIZE = 20 # Number of top individuals to carry over

# Define parameter ranges
ranges_dict = {col: {'Min': df[col].min(), 'Max': df[col].max()} for col in df.
    ↪ columns}
ranges_dict['NG FEED - Comp Mass Flow (KJ/h)']={'Min':50,'Max':100}
ranges_dict['Ratio (S/C)']={'Min':2.5,'Max':5}
ranges_dict['Reformer inlet temperature (°C)']={'Min':750,'Max':900}
ranges_dict['Reformer inlet pressure (kPa)']={'Min':2000,'Max':3500}
ranges_dict['Reformer outlet temperature (°C)']={'Min':800,'Max':1000}
ranges_dict['HTS - Inlet temperature (°C)']={'Min':300,'Max':450}
ranges_dict['HTS - Inlet pressure (kPa)']={'Min':1500,'Max':3500}
ranges_dict['LTS - Inlet temperature (°C)']={'Min':200,'Max':300}
ranges_dict['LTS - Inlet pressure (kPa)']={'Min':1500,'Max':3000}
ranges_dict['Condensor inlet temperature (°C)']={'Min':100,'Max':212.8}
ranges_dict['Condensor inlet pressure (kPa)']={'Min':800,'Max':2000}
ranges_dict['Methanator inlet temperature (°C)']={'Min':250,'Max':400}
ranges_dict['Hydrogen inlet temperature (°C)']={'Min':250,'Max':450}
ranges_dict['Nitrogen inlet temperature (°C)']={'Min':250,'Max':450}

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ranges_dict['Nitrogen inlet pressure (kPa)']={'Min':1000,'Max':2500}
ranges_dict['Synthesis gas inlet temperature (°C)']={'Min':350,'Max':500}
ranges_dict['Synthesis gas inlet pressure (kPa)']={'Min':15000,'Max':20000}
ranges_dict['Separator inlet temperature (°C)']={'Min':-33,'Max':10}
ranges_dict['K-100 - Delta P (kPa)']={'Min':100,'Max':400}
ranges_dict['K-102 - Delta P (kPa)']={'Min':1000,'Max':2500}

# It will helps us to go from index to name
predictures_list =['total_energy','NG FEED - Comp Mass Flow (KJ/h)','Ratio (S/
↳C)','Reformer inlet temperature (°C)',
    'Reformer outlet temperature (°C)','HTS - Inlet temperature (°C)','LTS -
↳Inlet temperature (°C)','Condensor inlet temperature (°C)',
    'Condensor inlet pressure (kPa)','Methanator inlet temperature (°C)','LTS -
↳Inlet pressure (kPa)',
    'Hydrogen inlet temperature (°C)','Nitrogen inlet temperature (°C)','K-100
↳- Delta P (kPa)','Synthesis gas inlet temperature (°C)','Separator inlet
↳temperature (°C)',
    'K-102 - Delta P (kPa)','Synthesis gas inlet pressure (kPa)','Separator
↳inlet pressure (kPa)'
    , 'Ammonia - Comp Mass Flow (kg/h)']

#load the data
#S_models
modelS2 = joblib.load('/home/houssam/internship project/trained_models/S-2
↳stream temperature_Pipeline.pkl')
modelS3 = joblib.load('/home/houssam/internship project/trained_models/S-3
↳stream temperature_Pipeline.pkl')

#energy
model1 = joblib.load('/home/houssam/internship project/trained_models/
↳Mixed_Steam_Pipeline.pkl')
model2 = joblib.load('/home/houssam/internship project/trained_models/
↳Reformer_Energy_Pipeline.pkl')
model3 = joblib.load('/home/houssam/internship project/trained_models/Q-102
↳Energy_Pipeline.pkl')
model4 = joblib.load('/home/houssam/internship project/trained_models/E-1-
↳Energy_Pipeline.pkl')
model5 = joblib.load('/home/houssam/internship project/trained_models/E-2-
↳Energy_Pipeline.pkl')
model6 = joblib.load('/home/houssam/internship project/trained_models/E-3-
↳Energy_Pipeline.pkl')
model7 = joblib.load('/home/houssam/internship project/trained_models/E-4-
↳Energy_Pipeline.pkl')
model8 = joblib.load('/home/houssam/internship project/trained_models/
↳E-separator- Energy_Pipeline.pkl')

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#Ammonia Mass
model9 = joblib.load('/home/houssam/internship project/trained_models/ Ammonia_
↳ Comp Mass Flow_Pipeline.pkl')

#CO2 emission
model10 = joblib.load('/home/houssam/internship project/trained_models/CO2_
↳emissions-Comp Mass Flow_Pipeline.pkl')

# Load the trained model
'''
total_energy = X0
NG FEED - Comp Mass Flow (KJ/h) = X1
Ratio (S/C) = X2
Reformer inlet temperature (°C) = X3
Reformer outlet temperature (°C) = X4
HTS - Inlet temperature (°C) = X5
LTS - Inlet temperature (°C) = X6
Condensor inlet temperature (°C) = X7
Condensor inlet pressure (kPa) = X8
Methanator inlet temperature (°C) = X9

Mixed Steam- Comp Mass Flow (KJ/h) = model1.predict(X1,X2)
LTS - Inlet pressure (kPa) = X10
Hydrogen inlet temperature (°C) = X11
Nitrogen inlet temperature (°C) = X12
K-100 - Delta P (kPa) = X13
S-2 stream temperature (°C) = modelS2.predict(X11,X12,X13)
S-3 stream temperature (°C) = modelS3.predict(X1,X2,X10,X11,X12,X13)
Synthesis gas inlet temperature (°C) = X14
Separator inlet temperature (°C) = X15
K-102 - Delta P (kPa) = X16
Synthesis gas inlet pressure (kPa) = X17
Separator inlet pressure (kPa) = X18

'''
def
↳predict_energy(X1,X2,X3,X4,X5,X6,X7,X8,X9,X10,X11,X12,X13,X14,X15,model1,model2,model3,mode
↳
Total_energy = 0

Mixed_Steam = model1.predict([[X1,X2]])[0]
S2 = modelS2.predict([[X11,X12,X13]])[0]
S3 = modelS3.predict([[X11,X12,X13,Mixed_Steam,X10]])[0]
Total_energy += model2.predict([[X3,X4]])[0]

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Total_energy += 4227654.693147199*X3 - 1507124898.3797724
Total_energy += model3.predict([[X4,X5]])[0]
Total_energy += -3082293.5646714796*X5 + 1077098983.4823322
Total_energy += -2640932.047069013*X6 + 953513207.3016243
Total_energy += -2640920.0637122733*X6 + 559925580.9712701
Total_energy += -2318202.284356667*X7 + 509146085.55179334
Total_energy += -2318641.333250072*X7 + 20190755.60255745
Total_energy += 11485.991953742923*X8 - 3851589.240937948
Total_energy += 2750563.303329907*X9 - 107470876.00437343
Total_energy += -2750561.059656047*X9 + 770251086.4382389
Total_energy += model4.predict([[Mixed_Steam,X10,X11,X13,S2]])[0]
Total_energy += model5.predict([[S2,S3,X13]])[0]
Total_energy += model6.predict([[Mixed_Steam,X10,X14]])[0]
Total_energy += -2579.93651444447*X14 + 747019.723192719
Total_energy += model7.predict([[Mixed_Steam, X10,X15]])[0]
Total_energy += model8.predict([[X15]])[0]

return Total_energy

def predict_Mass(X1,X2,X10,X11,X12,X13,X16,X17,X18,model1,model9):
    Mixed_Steam = model1.predict([[X1,X2]])[0]
    S2 = modelS2.predict([[X11,X12,X13]])[0]
    S3 = modelS3.predict([[X11,X12,X13,Mixed_Steam,X10]])[0]
    output_mass = model9.predict([[Mixed_Steam,X10,S2,S3,X13,X16,X17,X18]])[0]
    return output_mass

def CO2_emission(X1,X2,X10,model1,model10):
    Mixed_Steam = model1.predict([[X1,X2]])[0]
    output = model10.predict([[Mixed_Steam,X10]])[0]
    return output

def create_population(size,predictures_list):
    population = []
    for i in range(size):
        ind = []
        for prediction in predictions_list:
            Max = ranges_dict[prediction]['Max']
            Min = ranges_dict[prediction]['Min']
            ind.append(random.uniform(Min,Max))
        population.append(ind)
    return population

def fitness_function(individual):

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        predicted_mass = □
        ↪predict_Mass(individual[1],individual[2],individual[10],individual[11],individual[12],indiv
        predicted_CO2_mass = □
        ↪CO2_emission(individual[1],individual[2],individual[10],model1,model10)
        if predicted_mass > 164 and 172 > predicted_mass and predicted_CO2_mass □
        ↪< 256:
            predicted_energy = □
            ↪predict_energy(individual[1],individual[2],individual[3],individual[4],individual[5],indivi
            return -predicted_energy
        else:
            return -float('inf')

def tournament_selection(population, fitnesses, k=3):
    selected = random.sample(list(zip(population, fitnesses)), k)
    return min(selected, key=lambda item: item[1])[0]

def crossover(parent1, parent2):
    crossover_point = random.randint(2, len(parent1) - 2)
    child1 = parent1[:crossover_point] + parent2[crossover_point:]
    child2 = parent2[:crossover_point] + parent1[crossover_point:]
    return child1, child2

def mutate(individual):
    """Mutate an individual's parameters."""
    if random.random() < MUTATION_RATE:
        mutation_index = random.randint(1, len(individual) - 2)
        Max = ranges_dict[predictures_list[mutation_index]]['Max']
        Min = ranges_dict[predictures_list[mutation_index]]['Min']
        individual[mutation_index] = random.uniform(Min,Max)
    return individual

#the main algo

# Create initial population (type: df)
population = create_population(POPULATION_SIZE,predictures_list)

for generation in range(NUM_GENERATIONS):
    # Evaluate fitness of each individual

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fitnesses = [fitness_function(ind) for ind in population]

# Sort the population by fitness and preserve the top individuals
sorted_population = [ind for ind, fit in sorted(zip(population, fitnesses),
key=lambda x: x[1], reverse=True)]
elites = sorted_population[:ELITISM_SIZE]
elites_fitnesses = [fitness_function(ind) for ind in elites]

# Create a new population through selection, crossover, and mutation
new_population = elites # Start with the best individuals

while len(new_population) < POPULATION_SIZE:
    parent1 = tournament_selection(elites, elites_fitnesses)
    parent2 = tournament_selection(elites, elites_fitnesses)
    child1, child2 = crossover(parent1, parent2)
    new_population.append(mutate(child1))
    if len(new_population) < POPULATION_SIZE:
        new_population.append(mutate(child2))

# Replace the old population with the new population
population = new_population

# Print the best individual of the current generation
best_individual = max(population, key=fitness_function)
best_fitness = fitness_function(best_individual)
print(f'Generation {generation}: Best Fitness = {best_fitness}, Best
Individual = {best_individual}')

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Generation 0: Best Fitness = -4008856430.038673, Best Individual =
[5481627718.3384285, 61.56653406589433, 3.531675892029438, 751.2033748905658,
826.6660306340941, 440.6854444634479, 299.550586874092, 191.03735863257634,
1898.5286425941194, 344.2416479982271, 2743.3497651727303, 435.31517231189366,
381.55594498795335, 226.8498964001579, 464.61843895251457, -23.62600283077151,
1284.011501372542, 17702.142601154388, 10740.056316912021, 162.30280169086583]
Generation 1: Best Fitness = -4006894016.8862867, Best Individual =
[5507562689.514318, 85.80372261210611, 3.109707462604793, 807.0194519656526,
854.6110855428284, 443.2823094545564, 264.1281092173366, 202.90294472228962,
1913.648930654323, 298.3637420477382, 1856.8424174231461, 402.62077109149766,
284.4424312877771, 305.18610733084654, 467.44501847930553, -13.351846073095494,
1306.0658765301735, 17361.732005162565, 11471.44808067817, 168.80288784626936]
Generation 2: Best Fitness = -4006894016.8862867, Best Individual =

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[5507562689.514318, 85.80372261210611, 3.109707462604793, 807.0194519656526, 854.6110855428284, 443.2823094545564, 264.1281092173366, 202.90294472228962, 1913.648930654323, 298.3637420477382, 1856.8424174231461, 402.62077109149766, 284.4424312877771, 305.18610733084654, 467.44501847930553, -13.351846073095494, 1306.0658765301735, 17361.732005162565, 11471.44808067817, 168.80288784626936]

Generation 3: Best Fitness = -4005411661.7849264, Best Individual = [6389828973.746783, 77.86669027872068, 3.170284051859376, 852.779936909626, 826.675961033958, 400.0423331142466, 287.4408652671633, 201.07266892165845, 1445.1464309382966, 372.92666294321805, 2689.3432902049144, 279.79801976553193, 305.3428227986159, 248.62034604351666, 421.16311515605423, -26.289021283086427, 1444.13142870381, 19437.319111257224, 11794.524758568752, 170.32493319035356]

Generation 4: Best Fitness = -3999167259.394475, Best Individual = [6389828973.746783, 77.86669027872068, 3.170284051859376, 852.779936909626, 841.8070603491052, 438.3814225762825, 269.39994145991824, 208.7639126876776, 1836.5521922228227, 336.381370349147, 1862.936169787827, 438.6132111802663, 385.76722191678846, 248.62034604351666, 421.16311515605423, -26.289021283086427, 1444.13142870381, 19437.319111257224, 11794.524758568752, 170.32493319035356]

Generation 5: Best Fitness = -3996729017.7510934, Best Individual = [6358239491.627127, 71.65975140795294, 4.760871623595062, 801.2798350949355, 848.8426013387915, 419.1987633859297, 284.3725260798868, 206.30632897252576, 1836.5521922228227, 369.99662863383975, 2733.5387438508196, 325.1763353413239, 412.80619798069614, 151.18749649970653, 410.54389875361113, -1.0437374874189018, 1040.0840014608982, 19392.528882023224, 11571.868579538332, 173.2448185714548]

Generation 6: Best Fitness = -3996729017.7510934, Best Individual = [6358239491.627127, 71.65975140795294, 4.760871623595062, 801.2798350949355, 848.8426013387915, 419.1987633859297, 284.3725260798868, 206.30632897252576, 1836.5521922228227, 369.99662863383975, 2733.5387438508196, 325.1763353413239, 412.80619798069614, 151.18749649970653, 410.54389875361113, -1.0437374874189018, 1040.0840014608982, 19392.528882023224, 11571.868579538332, 173.2448185714548]

Generation 7: Best Fitness = -3979783641.98723, Best Individual = [4991254306.353332, 74.0634129953899, 4.074298939329507, 836.2829011391914, 833.2951915184516, 398.77297961336654, 277.76620424041977, 208.7639126876776, 809.9592598992036, 287.8998354200334, 2227.626681016345, 329.8002987381965, 332.6250135840495, 372.45544494685583, 410.54389875361113, -1.0437374874189018, 1040.0840014608982, 19392.528882023224, 10647.782663216989, 173.2448185714548]

Generation 8: Best Fitness = -3881388318.7592044, Best Individual = [5037097809.867791, 61.589067992957105, 4.4208823361211245, 787.0979572837067, 841.28283530473, 435.8965437590683, 299.6115478522903, 208.7639126876776, 952.2779768412478, 380.67096572668913, 2733.5387438508196, 255.13349387043766, 412.80619798069614, 151.18749649970653, 399.1933604271219, -18.538112467281096, 2134.868424647361, 18905.3593927836, 11264.129055270696, 166.61520563923494]

Generation 9: Best Fitness = -3881388318.7592044, Best Individual = [5037097809.867791, 61.589067992957105, 4.4208823361211245, 787.0979572837067, 841.28283530473, 435.8965437590683, 299.6115478522903, 208.7639126876776, 952.2779768412478, 380.67096572668913, 2733.5387438508196, 255.13349387043766, 412.80619798069614, 151.18749649970653, 399.1933604271219, -18.538112467281096, 2134.868424647361, 18905.3593927836, 11264.129055270696, 166.61520563923494]

Generation 10: Best Fitness = -3875248523.4098425, Best Individual =

[4991254306.353332, 74.0634129953899, 4.074298939329507, 801.2798350949355, 833.2951915184516, 438.3814225762825, 292.0711684855221, 208.7639126876776, 952.2779768412478, 390.5780601825915, 1856.8424174231461, 438.6132111802663, 385.76722191678846, 256.1609399551684, 475.3286639163187, -1.0437374874189018, 1040.0840014608982, 19392.528882023224, 10447.430650796432, 173.2448185714548]

Generation 11: Best Fitness = -3802071245.7552085, Best Individual = [4991254306.353332, 74.0634129953899, 2.976321709052205, 836.6351615815163, 837.6795922745987, 438.3814225762825, 294.90577346491705, 208.7639126876776, 1836.5521922228227, 265.40658155815055, 1862.936169787827, 438.6132111802663, 385.76722191678846, 256.1609399551684, 475.3286639163187, -2.7554824938489126, 1040.0840014608982, 19392.528882023224, 10447.430650796432, 173.2448185714548]

Generation 12: Best Fitness = -3802071245.7552085, Best Individual = [4991254306.353332, 74.0634129953899, 2.976321709052205, 836.6351615815163, 837.6795922745987, 438.3814225762825, 294.90577346491705, 208.7639126876776, 1836.5521922228227, 265.40658155815055, 1862.936169787827, 438.6132111802663, 385.76722191678846, 256.1609399551684, 475.3286639163187, -2.7554824938489126, 1040.0840014608982, 19392.528882023224, 10447.430650796432, 173.2448185714548]

Generation 13: Best Fitness = -3802071245.7552085, Best Individual = [4991254306.353332, 74.0634129953899, 2.976321709052205, 836.6351615815163, 837.6795922745987, 438.3814225762825, 294.90577346491705, 208.7639126876776, 1836.5521922228227, 265.40658155815055, 1862.936169787827, 438.6132111802663, 385.76722191678846, 256.1609399551684, 475.3286639163187, -2.7554824938489126, 1040.0840014608982, 19392.528882023224, 10447.430650796432, 173.2448185714548]

Generation 14: Best Fitness = -3802071245.7552085, Best Individual = [4991254306.353332, 74.0634129953899, 2.976321709052205, 836.6351615815163, 837.6795922745987, 438.3814225762825, 294.90577346491705, 208.7639126876776, 1836.5521922228227, 265.40658155815055, 1862.936169787827, 438.6132111802663, 385.76722191678846, 256.1609399551684, 475.3286639163187, -2.7554824938489126, 1040.0840014608982, 19392.528882023224, 10447.430650796432, 173.2448185714548]

Generation 15: Best Fitness = -3802071245.7552085, Best Individual = [4991254306.353332, 74.0634129953899, 2.976321709052205, 836.6351615815163, 837.6795922745987, 438.3814225762825, 294.90577346491705, 208.7639126876776, 1836.5521922228227, 265.40658155815055, 1862.936169787827, 438.6132111802663, 385.76722191678846, 256.1609399551684, 475.3286639163187, -2.7554824938489126, 1040.0840014608982, 19392.528882023224, 10447.430650796432, 173.2448185714548]

Generation 16: Best Fitness = -3802071245.7552085, Best Individual = [4991254306.353332, 74.0634129953899, 2.976321709052205, 836.6351615815163, 837.6795922745987, 438.3814225762825, 294.90577346491705, 208.7639126876776, 1836.5521922228227, 265.40658155815055, 1862.936169787827, 438.6132111802663, 385.76722191678846, 256.1609399551684, 475.3286639163187, -2.7554824938489126, 1040.0840014608982, 19392.528882023224, 10447.430650796432, 173.2448185714548]

Generation 17: Best Fitness = -3802071245.7552085, Best Individual = [4991254306.353332, 74.0634129953899, 2.976321709052205, 836.6351615815163, 837.6795922745987, 438.3814225762825, 294.90577346491705, 208.7639126876776, 1836.5521922228227, 265.40658155815055, 1862.936169787827, 438.6132111802663, 385.76722191678846, 256.1609399551684, 475.3286639163187, -2.7554824938489126, 1040.0840014608982, 19392.528882023224, 10447.430650796432, 173.2448185714548]

Generation 18: Best Fitness = -3802071245.7552085, Best Individual =

[4991254306.353332, 74.0634129953899, 2.976321709052205, 836.6351615815163, 837.6795922745987, 438.3814225762825, 294.90577346491705, 208.7639126876776, 1836.5521922228227, 265.40658155815055, 1862.936169787827, 438.6132111802663, 385.76722191678846, 256.1609399551684, 475.3286639163187, -2.7554824938489126, 1040.0840014608982, 19392.528882023224, 10447.430650796432, 173.2448185714548]

Generation 19: Best Fitness = -3798405861.479123, Best Individual = [5507562689.514318, 85.80372261210611, 3.109707462604793, 807.0194519656526, 837.6795922745987, 438.3814225762825, 299.550586874092, 208.7639126876776, 952.2779768412478, 390.5780601825915, 1856.8424174231461, 438.6132111802663, 385.76722191678846, 233.85689478597567, 475.3286639163187, -1.0437374874189018, 1040.0840014608982, 19392.528882023224, 10447.430650796432, 173.2448185714548]

Generation 20: Best Fitness = -3798405861.479123, Best Individual = [5507562689.514318, 85.80372261210611, 3.109707462604793, 807.0194519656526, 837.6795922745987, 438.3814225762825, 299.550586874092, 208.7639126876776, 952.2779768412478, 390.5780601825915, 1856.8424174231461, 438.6132111802663, 385.76722191678846, 233.85689478597567, 475.3286639163187, -1.0437374874189018, 1040.0840014608982, 19392.528882023224, 10447.430650796432, 173.2448185714548]

Generation 21: Best Fitness = -3791604552.067683, Best Individual = [4991254306.353332, 74.0634129953899, 2.9379569924920172, 836.4879029331403, 837.6795922745987, 438.3814225762825, 294.90577346491705, 208.7639126876776, 947.5101432000857, 365.7965128300012, 1929.129950377941, 438.6132111802663, 385.76722191678846, 256.1609399551684, 475.3286639163187, -1.0437374874189018, 1040.0840014608982, 19392.528882023224, 10447.430650796432, 173.2448185714548]

Generation 22: Best Fitness = -3789779514.178058, Best Individual = [4991254306.353332, 74.0634129953899, 3.109707462604793, 807.0194519656526, 826.6660306340941, 440.6854444634479, 299.550586874092, 208.7639126876776, 952.2779768412478, 331.94296249553804, 1929.129950377941, 438.6132111802663, 385.76722191678846, 256.1609399551684, 475.3286639163187, -1.0437374874189018, 1040.0840014608982, 19392.528882023224, 10536.937811976079, 173.2448185714548]

Generation 23: Best Fitness = -3773374788.5966945, Best Individual = [5507562689.514318, 85.80372261210611, 3.109707462604793, 807.0194519656526, 837.6795922745987, 445.93115872452785, 299.77253657808063, 208.7639126876776, 904.0798866541297, 381.40713792172403, 2208.03566170745, 438.6132111802663, 385.76722191678846, 256.1609399551684, 475.3286639163187, -1.0437374874189018, 1040.0840014608982, 19392.528882023224, 10447.430650796432, 173.2448185714548]

Generation 24: Best Fitness = -3773374788.5966945, Best Individual = [5507562689.514318, 85.80372261210611, 3.109707462604793, 807.0194519656526, 837.6795922745987, 445.93115872452785, 299.77253657808063, 208.7639126876776, 904.0798866541297, 381.40713792172403, 2208.03566170745, 438.6132111802663, 385.76722191678846, 256.1609399551684, 475.3286639163187, -1.0437374874189018, 1040.0840014608982, 19392.528882023224, 10447.430650796432, 173.2448185714548]

Generation 25: Best Fitness = -3756765682.482962, Best Individual = [4991254306.353332, 74.0634129953899, 2.9379569924920172, 836.4879029331403, 834.5660639888896, 438.3814225762825, 294.90577346491705, 208.7639126876776, 1046.6911836999682, 260.6660143637603, 2208.03566170745, 282.58967954455295, 385.76722191678846, 256.1609399551684, 475.3286639163187, -1.0437374874189018, 2134.868424647361, 18905.3593927836, 10740.056316912021, 162.30280169086583]

Generation 26: Best Fitness = -3724014908.0535364, Best Individual =

[4991254306.353332, 74.0634129953899, 2.9379569924920172, 836.4879029331403, 826.6660306340941, 440.6854444634479, 299.550586874092, 208.7639126876776, 952.2779768412478, 331.94296249553804, 1929.129950377941, 438.6132111802663, 381.33533355236574, 256.1609399551684, 475.3286639163187, -1.0437374874189018, 2134.868424647361, 18905.3593927836, 10740.056316912021, 162.30280169086583]

Generation 27: Best Fitness = -3724014908.0535364, Best Individual = [4991254306.353332, 74.0634129953899, 2.9379569924920172, 836.4879029331403, 826.6660306340941, 440.6854444634479, 299.550586874092, 208.7639126876776, 952.2779768412478, 331.94296249553804, 1929.129950377941, 438.6132111802663, 381.33533355236574, 256.1609399551684, 475.3286639163187, -1.0437374874189018, 2134.868424647361, 18905.3593927836, 10740.056316912021, 162.30280169086583]

Generation 28: Best Fitness = -3724014908.0535364, Best Individual = [4991254306.353332, 74.0634129953899, 2.9379569924920172, 836.4879029331403, 826.6660306340941, 440.6854444634479, 299.550586874092, 208.7639126876776, 952.2779768412478, 331.94296249553804, 1929.129950377941, 438.6132111802663, 381.33533355236574, 256.1609399551684, 475.3286639163187, -1.0437374874189018, 2134.868424647361, 18905.3593927836, 10740.056316912021, 162.30280169086583]

Generation 29: Best Fitness = -3724014908.0535364, Best Individual = [4991254306.353332, 74.0634129953899, 2.9379569924920172, 836.4879029331403, 826.6660306340941, 440.6854444634479, 299.550586874092, 208.7639126876776, 952.2779768412478, 331.94296249553804, 1929.129950377941, 438.6132111802663, 381.33533355236574, 256.1609399551684, 475.3286639163187, -1.0437374874189018, 2134.868424647361, 18905.3593927836, 10740.056316912021, 162.30280169086583]

Generation 30: Best Fitness = -3724014908.0535364, Best Individual = [4991254306.353332, 74.0634129953899, 2.9379569924920172, 836.4879029331403, 826.6660306340941, 440.6854444634479, 299.550586874092, 208.7639126876776, 952.2779768412478, 331.94296249553804, 1929.129950377941, 438.6132111802663, 381.33533355236574, 256.1609399551684, 475.3286639163187, -1.0437374874189018, 2134.868424647361, 18905.3593927836, 10740.056316912021, 162.30280169086583]

Generation 31: Best Fitness = -3724014908.0535364, Best Individual = [4991254306.353332, 74.0634129953899, 2.9379569924920172, 836.4879029331403, 826.6660306340941, 440.6854444634479, 299.550586874092, 208.7639126876776, 952.2779768412478, 331.94296249553804, 1929.129950377941, 438.6132111802663, 381.33533355236574, 256.1609399551684, 475.3286639163187, -1.0437374874189018, 2134.868424647361, 18905.3593927836, 10740.056316912021, 162.30280169086583]

Generation 32: Best Fitness = -3724014908.0535364, Best Individual = [4991254306.353332, 74.0634129953899, 2.9379569924920172, 836.4879029331403, 826.6660306340941, 440.6854444634479, 299.550586874092, 208.7639126876776, 952.2779768412478, 331.94296249553804, 1929.129950377941, 438.6132111802663, 381.33533355236574, 256.1609399551684, 475.3286639163187, -1.0437374874189018, 2134.868424647361, 18905.3593927836, 10740.056316912021, 162.30280169086583]

Generation 33: Best Fitness = -3708070413.0820136, Best Individual = [4991254306.353332, 74.0634129953899, 2.9379569924920172, 836.4879029331403, 834.5660639888896, 445.93115872452785, 299.550586874092, 208.7639126876776, 952.2779768412478, 331.94296249553804, 1929.129950377941, 328.7165128679039, 385.76722191678846, 256.1609399551684, 437.06410895684127, -1.0437374874189018, 1040.0840014608982, 19392.528882023224, 10447.430650796432, 173.2448185714548]

Generation 34: Best Fitness = -3706849095.757524, Best Individual =

[4991254306.353332, 74.0634129953899, 2.976321709052205, 836.6351615815163, 815.8859896135058, 445.93115872452785, 299.77253657808063, 208.7639126876776, 947.5101432000857, 365.7965128300012, 1929.129950377941, 389.8884468878266, 385.76722191678846, 256.1609399551684, 475.3286639163187, -1.0437374874189018, 1431.87431509086, 18905.3593927836, 10608.696340517743, 173.2448185714548]

Generation 35: Best Fitness = -3706849095.757524, Best Individual = [4991254306.353332, 74.0634129953899, 2.976321709052205, 836.6351615815163, 815.8859896135058, 445.93115872452785, 299.77253657808063, 208.7639126876776, 947.5101432000857, 365.7965128300012, 1929.129950377941, 389.8884468878266, 385.76722191678846, 256.1609399551684, 475.3286639163187, -1.0437374874189018, 1431.87431509086, 18905.3593927836, 10608.696340517743, 173.2448185714548]

Generation 36: Best Fitness = -3704987019.801774, Best Individual = [4991254306.353332, 74.0634129953899, 2.9379569924920172, 836.4879029331403, 834.5660639888896, 445.93115872452785, 299.77253657808063, 208.7639126876776, 812.9044946001017, 265.40658155815055, 2208.03566170745, 389.8884468878266, 260.63391364642155, 256.1609399551684, 475.3286639163187, 6.132285250540143, 1040.0840014608982, 19392.528882023224, 10447.430650796432, 173.2448185714548]

Generation 37: Best Fitness = -3704987019.801774, Best Individual = [4991254306.353332, 74.0634129953899, 2.9379569924920172, 836.4879029331403, 834.5660639888896, 445.93115872452785, 299.77253657808063, 208.7639126876776, 812.9044946001017, 265.40658155815055, 2208.03566170745, 389.8884468878266, 260.63391364642155, 256.1609399551684, 475.3286639163187, 6.132285250540143, 1040.0840014608982, 19392.528882023224, 10447.430650796432, 173.2448185714548]

Generation 38: Best Fitness = -3704987019.801774, Best Individual = [4991254306.353332, 74.0634129953899, 2.9379569924920172, 836.4879029331403, 834.5660639888896, 445.93115872452785, 299.77253657808063, 208.7639126876776, 812.9044946001017, 265.40658155815055, 2208.03566170745, 389.8884468878266, 260.63391364642155, 256.1609399551684, 475.3286639163187, 6.132285250540143, 1040.0840014608982, 19392.528882023224, 10447.430650796432, 173.2448185714548]

Generation 39: Best Fitness = -3704987019.801774, Best Individual = [4991254306.353332, 74.0634129953899, 2.9379569924920172, 836.4879029331403, 834.5660639888896, 445.93115872452785, 299.77253657808063, 208.7639126876776, 812.9044946001017, 265.40658155815055, 2208.03566170745, 389.8884468878266, 260.63391364642155, 256.1609399551684, 475.3286639163187, 6.132285250540143, 1040.0840014608982, 19392.528882023224, 10447.430650796432, 173.2448185714548]

Generation 40: Best Fitness = -3704987019.801774, Best Individual = [4991254306.353332, 74.0634129953899, 2.9379569924920172, 836.4879029331403, 834.5660639888896, 445.93115872452785, 299.77253657808063, 208.7639126876776, 812.9044946001017, 265.40658155815055, 2208.03566170745, 389.8884468878266, 260.63391364642155, 256.1609399551684, 475.3286639163187, 6.132285250540143, 1040.0840014608982, 19392.528882023224, 10447.430650796432, 173.2448185714548]

Generation 41: Best Fitness = -3704987019.801774, Best Individual = [4991254306.353332, 74.0634129953899, 2.9379569924920172, 836.4879029331403, 834.5660639888896, 445.93115872452785, 299.77253657808063, 208.7639126876776, 812.9044946001017, 265.40658155815055, 2208.03566170745, 389.8884468878266, 260.63391364642155, 256.1609399551684, 475.3286639163187, 6.132285250540143, 1040.0840014608982, 19392.528882023224, 10447.430650796432, 173.2448185714548]

Generation 42: Best Fitness = -3704987019.801774, Best Individual =

[4991254306.353332, 74.0634129953899, 2.9379569924920172, 836.4879029331403, 834.5660639888896, 445.93115872452785, 299.77253657808063, 208.7639126876776, 812.9044946001017, 265.40658155815055, 2208.03566170745, 389.8884468878266, 260.63391364642155, 256.1609399551684, 475.3286639163187, 6.132285250540143, 1040.0840014608982, 19392.528882023224, 10447.430650796432, 173.2448185714548]
Generation 43: Best Fitness = -3704987019.801774, Best Individual =
[4991254306.353332, 74.0634129953899, 2.9379569924920172, 836.4879029331403, 834.5660639888896, 445.93115872452785, 299.77253657808063, 208.7639126876776, 812.9044946001017, 265.40658155815055, 2208.03566170745, 389.8884468878266, 260.63391364642155, 256.1609399551684, 475.3286639163187, 6.132285250540143, 1040.0840014608982, 19392.528882023224, 10447.430650796432, 173.2448185714548]
Generation 44: Best Fitness = -3704987019.801774, Best Individual =
[4991254306.353332, 74.0634129953899, 2.9379569924920172, 836.4879029331403, 834.5660639888896, 445.93115872452785, 299.77253657808063, 208.7639126876776, 812.9044946001017, 265.40658155815055, 2208.03566170745, 389.8884468878266, 260.63391364642155, 256.1609399551684, 475.3286639163187, 6.132285250540143, 1040.0840014608982, 19392.528882023224, 10447.430650796432, 173.2448185714548]
Generation 45: Best Fitness = -3697792932.2818546, Best Individual =
[4991254306.353332, 74.0634129953899, 2.9379569924920172, 836.4879029331403, 834.5660639888896, 445.93115872452785, 299.77253657808063, 211.0491929813066, 1046.6911836999682, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 256.1609399551684, 359.1190537110743, -1.0437374874189018, 1382.3798932062741, 19392.528882023224, 11377.144627159429, 173.2448185714548]
Generation 46: Best Fitness = -3697792932.2818546, Best Individual =
[4991254306.353332, 74.0634129953899, 2.9379569924920172, 836.4879029331403, 834.5660639888896, 445.93115872452785, 299.77253657808063, 211.0491929813066, 1046.6911836999682, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 256.1609399551684, 359.1190537110743, -1.0437374874189018, 1382.3798932062741, 19392.528882023224, 11377.144627159429, 173.2448185714548]
Generation 47: Best Fitness = -3697792932.2818546, Best Individual =
[4991254306.353332, 74.0634129953899, 2.9379569924920172, 836.4879029331403, 834.5660639888896, 445.93115872452785, 299.77253657808063, 211.0491929813066, 1046.6911836999682, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 256.1609399551684, 359.1190537110743, -1.0437374874189018, 1382.3798932062741, 19392.528882023224, 11377.144627159429, 173.2448185714548]
Generation 48: Best Fitness = -3697792932.2818546, Best Individual =
[4991254306.353332, 74.0634129953899, 2.9379569924920172, 836.4879029331403, 834.5660639888896, 445.93115872452785, 299.77253657808063, 211.0491929813066, 1046.6911836999682, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 256.1609399551684, 359.1190537110743, -1.0437374874189018, 1382.3798932062741, 19392.528882023224, 11377.144627159429, 173.2448185714548]
Generation 49: Best Fitness = -3697792932.2818546, Best Individual =
[4991254306.353332, 74.0634129953899, 2.9379569924920172, 836.4879029331403, 834.5660639888896, 445.93115872452785, 299.77253657808063, 211.0491929813066, 1046.6911836999682, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 256.1609399551684, 359.1190537110743, -1.0437374874189018, 1382.3798932062741, 19392.528882023224, 11377.144627159429, 173.2448185714548]
Generation 50: Best Fitness = -3697792932.2818546, Best Individual =

[4991254306.353332, 74.0634129953899, 2.9379569924920172, 836.4879029331403, 834.5660639888896, 445.93115872452785, 299.77253657808063, 211.0491929813066, 1046.6911836999682, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 256.1609399551684, 359.1190537110743, -1.0437374874189018, 1382.3798932062741, 19392.528882023224, 11377.144627159429, 173.2448185714548]
Generation 51: Best Fitness = -3697792932.2818546, Best Individual =
[4991254306.353332, 74.0634129953899, 2.9379569924920172, 836.4879029331403, 834.5660639888896, 445.93115872452785, 299.77253657808063, 211.0491929813066, 1046.6911836999682, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 256.1609399551684, 359.1190537110743, -1.0437374874189018, 1382.3798932062741, 19392.528882023224, 11377.144627159429, 173.2448185714548]
Generation 52: Best Fitness = -3697792932.2818546, Best Individual =
[4991254306.353332, 74.0634129953899, 2.9379569924920172, 836.4879029331403, 834.5660639888896, 445.93115872452785, 299.77253657808063, 211.0491929813066, 1046.6911836999682, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 256.1609399551684, 359.1190537110743, -1.0437374874189018, 1382.3798932062741, 19392.528882023224, 11377.144627159429, 173.2448185714548]
Generation 53: Best Fitness = -3697792932.2818546, Best Individual =
[4991254306.353332, 74.0634129953899, 2.9379569924920172, 836.4879029331403, 834.5660639888896, 445.93115872452785, 299.77253657808063, 211.0491929813066, 1046.6911836999682, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 256.1609399551684, 359.1190537110743, -1.0437374874189018, 1382.3798932062741, 19392.528882023224, 11377.144627159429, 173.2448185714548]
Generation 54: Best Fitness = -3697354933.362265, Best Individual =
[4991254306.353332, 63.41236421269042, 3.2955705104130484, 836.4879029331403, 834.5660639888896, 445.93115872452785, 299.550586874092, 211.02307897180174, 952.2779768412478, 331.94296249553804, 1929.129950377941, 328.7165128679039, 342.15755604153884, 256.1609399551684, 475.3286639163187, -1.0437374874189018, 1382.3798932062741, 16551.76851963386, 10425.792969363802, 173.2448185714548]
Generation 55: Best Fitness = -3697354933.362265, Best Individual =
[4991254306.353332, 63.41236421269042, 3.2955705104130484, 836.4879029331403, 834.5660639888896, 445.93115872452785, 299.550586874092, 211.02307897180174, 952.2779768412478, 331.94296249553804, 1929.129950377941, 328.7165128679039, 342.15755604153884, 256.1609399551684, 475.3286639163187, -1.0437374874189018, 1382.3798932062741, 16551.76851963386, 10425.792969363802, 173.2448185714548]
Generation 56: Best Fitness = -3697354933.362265, Best Individual =
[4991254306.353332, 63.41236421269042, 3.2955705104130484, 836.4879029331403, 834.5660639888896, 445.93115872452785, 299.550586874092, 211.02307897180174, 952.2779768412478, 331.94296249553804, 1929.129950377941, 328.7165128679039, 342.15755604153884, 256.1609399551684, 475.3286639163187, -1.0437374874189018, 1382.3798932062741, 16551.76851963386, 10425.792969363802, 173.2448185714548]
Generation 57: Best Fitness = -3697354933.362265, Best Individual =
[4991254306.353332, 63.41236421269042, 3.2955705104130484, 836.4879029331403, 834.5660639888896, 445.93115872452785, 299.550586874092, 211.02307897180174, 952.2779768412478, 331.94296249553804, 1929.129950377941, 328.7165128679039, 342.15755604153884, 256.1609399551684, 475.3286639163187, -1.0437374874189018, 1382.3798932062741, 16551.76851963386, 10425.792969363802, 173.2448185714548]
Generation 58: Best Fitness = -3697354933.362265, Best Individual =

[4991254306.353332, 63.41236421269042, 3.2955705104130484, 836.4879029331403, 834.5660639888896, 445.93115872452785, 299.550586874092, 211.02307897180174, 952.2779768412478, 331.94296249553804, 1929.129950377941, 328.7165128679039, 342.15755604153884, 256.1609399551684, 475.3286639163187, -1.0437374874189018, 1382.3798932062741, 16551.76851963386, 10425.792969363802, 173.2448185714548]

Generation 59: Best Fitness = -3697354933.362265, Best Individual =

[4991254306.353332, 63.41236421269042, 3.2955705104130484, 836.4879029331403, 834.5660639888896, 445.93115872452785, 299.550586874092, 211.02307897180174, 952.2779768412478, 331.94296249553804, 1929.129950377941, 328.7165128679039, 342.15755604153884, 256.1609399551684, 475.3286639163187, -1.0437374874189018, 1382.3798932062741, 16551.76851963386, 10425.792969363802, 173.2448185714548]

Generation 60: Best Fitness = -3697354933.362265, Best Individual =

[4991254306.353332, 63.41236421269042, 3.2955705104130484, 836.4879029331403, 834.5660639888896, 445.93115872452785, 299.550586874092, 211.02307897180174, 952.2779768412478, 331.94296249553804, 1929.129950377941, 328.7165128679039, 342.15755604153884, 256.1609399551684, 475.3286639163187, -1.0437374874189018, 1382.3798932062741, 16551.76851963386, 10425.792969363802, 173.2448185714548]

Generation 61: Best Fitness = -3697354933.362265, Best Individual =

[4991254306.353332, 63.41236421269042, 3.2955705104130484, 836.4879029331403, 834.5660639888896, 445.93115872452785, 299.550586874092, 211.02307897180174, 952.2779768412478, 331.94296249553804, 1929.129950377941, 328.7165128679039, 342.15755604153884, 256.1609399551684, 475.3286639163187, -1.0437374874189018, 1382.3798932062741, 16551.76851963386, 10425.792969363802, 173.2448185714548]

Generation 62: Best Fitness = -3697354933.362265, Best Individual =

[4991254306.353332, 63.41236421269042, 3.2955705104130484, 836.4879029331403, 834.5660639888896, 445.93115872452785, 299.550586874092, 211.02307897180174, 952.2779768412478, 331.94296249553804, 1929.129950377941, 328.7165128679039, 342.15755604153884, 256.1609399551684, 475.3286639163187, -1.0437374874189018, 1382.3798932062741, 16551.76851963386, 10425.792969363802, 173.2448185714548]

Generation 63: Best Fitness = -3697354933.362265, Best Individual =

[4991254306.353332, 63.41236421269042, 3.2955705104130484, 836.4879029331403, 834.5660639888896, 445.93115872452785, 299.550586874092, 211.02307897180174, 952.2779768412478, 331.94296249553804, 1929.129950377941, 328.7165128679039, 342.15755604153884, 256.1609399551684, 475.3286639163187, -1.0437374874189018, 1382.3798932062741, 16551.76851963386, 10425.792969363802, 173.2448185714548]

Generation 64: Best Fitness = -3692028666.7495565, Best Individual =

[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.5063774383783, 299.77253657808063, 208.7639126876776, 812.9044946001017, 265.40658155815055, 2208.03566170745, 446.3490908361761, 260.63391364642155, 256.1609399551684, 492.894860112155, 9.046498163007648, 1382.3798932062741, 16551.76851963386, 10447.430650796432, 173.2448185714548]

Generation 65: Best Fitness = -3692028666.7495565, Best Individual =

[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.5063774383783, 299.77253657808063, 208.7639126876776, 812.9044946001017, 265.40658155815055, 2208.03566170745, 446.3490908361761, 260.63391364642155, 256.1609399551684, 492.894860112155, 9.046498163007648, 1382.3798932062741, 16551.76851963386, 10447.430650796432, 173.2448185714548]

Generation 66: Best Fitness = -3692028666.7495565, Best Individual =

[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.5063774383783, 299.77253657808063, 208.7639126876776, 812.9044946001017, 265.40658155815055, 2208.03566170745, 446.3490908361761, 260.63391364642155, 256.1609399551684, 492.894860112155, 9.046498163007648, 1382.3798932062741, 16551.76851963386, 10447.430650796432, 173.2448185714548]
Generation 67: Best Fitness = -3692028666.7495565, Best Individual =
[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.5063774383783, 299.77253657808063, 208.7639126876776, 812.9044946001017, 265.40658155815055, 2208.03566170745, 446.3490908361761, 260.63391364642155, 256.1609399551684, 492.894860112155, 9.046498163007648, 1382.3798932062741, 16551.76851963386, 10447.430650796432, 173.2448185714548]
Generation 68: Best Fitness = -3692028666.7495565, Best Individual =
[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.5063774383783, 299.77253657808063, 208.7639126876776, 812.9044946001017, 265.40658155815055, 2208.03566170745, 446.3490908361761, 260.63391364642155, 256.1609399551684, 492.894860112155, 9.046498163007648, 1382.3798932062741, 16551.76851963386, 10447.430650796432, 173.2448185714548]
Generation 69: Best Fitness = -3692028666.7495565, Best Individual =
[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.5063774383783, 299.77253657808063, 208.7639126876776, 812.9044946001017, 265.40658155815055, 2208.03566170745, 446.3490908361761, 260.63391364642155, 256.1609399551684, 492.894860112155, 9.046498163007648, 1382.3798932062741, 16551.76851963386, 10447.430650796432, 173.2448185714548]
Generation 70: Best Fitness = -3692028666.7495565, Best Individual =
[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.5063774383783, 299.77253657808063, 208.7639126876776, 812.9044946001017, 265.40658155815055, 2208.03566170745, 446.3490908361761, 260.63391364642155, 256.1609399551684, 492.894860112155, 9.046498163007648, 1382.3798932062741, 16551.76851963386, 10447.430650796432, 173.2448185714548]
Generation 71: Best Fitness = -3692028666.7495565, Best Individual =
[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.5063774383783, 299.77253657808063, 208.7639126876776, 812.9044946001017, 265.40658155815055, 2208.03566170745, 446.3490908361761, 260.63391364642155, 256.1609399551684, 492.894860112155, 9.046498163007648, 1382.3798932062741, 16551.76851963386, 10447.430650796432, 173.2448185714548]
Generation 72: Best Fitness = -3692028666.7495565, Best Individual =
[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.5063774383783, 299.77253657808063, 208.7639126876776, 812.9044946001017, 265.40658155815055, 2208.03566170745, 446.3490908361761, 260.63391364642155, 256.1609399551684, 492.894860112155, 9.046498163007648, 1382.3798932062741, 16551.76851963386, 10447.430650796432, 173.2448185714548]
Generation 73: Best Fitness = -3692028666.7495565, Best Individual =
[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.5063774383783, 299.77253657808063, 208.7639126876776, 812.9044946001017, 265.40658155815055, 2208.03566170745, 446.3490908361761, 260.63391364642155, 256.1609399551684, 492.894860112155, 9.046498163007648, 1382.3798932062741, 16551.76851963386, 10447.430650796432, 173.2448185714548]
Generation 74: Best Fitness = -3692028666.7495565, Best Individual =

[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.5063774383783, 299.77253657808063, 208.7639126876776, 812.9044946001017, 265.40658155815055, 2208.03566170745, 446.3490908361761, 260.63391364642155, 256.1609399551684, 492.894860112155, 9.046498163007648, 1382.3798932062741, 16551.76851963386, 10447.430650796432, 173.2448185714548]

Generation 75: Best Fitness = -3692028666.7495565, Best Individual =

[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.5063774383783, 299.77253657808063, 208.7639126876776, 812.9044946001017, 265.40658155815055, 2208.03566170745, 446.3490908361761, 260.63391364642155, 256.1609399551684, 492.894860112155, 9.046498163007648, 1382.3798932062741, 16551.76851963386, 10447.430650796432, 173.2448185714548]

Generation 76: Best Fitness = -3692028666.7495565, Best Individual =

[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.5063774383783, 299.77253657808063, 208.7639126876776, 812.9044946001017, 265.40658155815055, 2208.03566170745, 446.3490908361761, 260.63391364642155, 256.1609399551684, 492.894860112155, 9.046498163007648, 1382.3798932062741, 16551.76851963386, 10447.430650796432, 173.2448185714548]

Generation 77: Best Fitness = -3692028666.7495565, Best Individual =

[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.5063774383783, 299.77253657808063, 208.7639126876776, 812.9044946001017, 265.40658155815055, 2208.03566170745, 446.3490908361761, 260.63391364642155, 256.1609399551684, 492.894860112155, 9.046498163007648, 1382.3798932062741, 16551.76851963386, 10447.430650796432, 173.2448185714548]

Generation 78: Best Fitness = -3692028666.7495565, Best Individual =

[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.5063774383783, 299.77253657808063, 208.7639126876776, 812.9044946001017, 265.40658155815055, 2208.03566170745, 446.3490908361761, 260.63391364642155, 256.1609399551684, 492.894860112155, 9.046498163007648, 1382.3798932062741, 16551.76851963386, 10447.430650796432, 173.2448185714548]

Generation 79: Best Fitness = -3692028666.7495565, Best Individual =

[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.5063774383783, 299.77253657808063, 208.7639126876776, 812.9044946001017, 265.40658155815055, 2208.03566170745, 446.3490908361761, 260.63391364642155, 256.1609399551684, 492.894860112155, 9.046498163007648, 1382.3798932062741, 16551.76851963386, 10447.430650796432, 173.2448185714548]

Generation 80: Best Fitness = -3692028666.7495565, Best Individual =

[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.5063774383783, 299.77253657808063, 208.7639126876776, 812.9044946001017, 265.40658155815055, 2208.03566170745, 446.3490908361761, 260.63391364642155, 256.1609399551684, 492.894860112155, 9.046498163007648, 1382.3798932062741, 16551.76851963386, 10447.430650796432, 173.2448185714548]

Generation 81: Best Fitness = -3692028666.7495565, Best Individual =

[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.5063774383783, 299.77253657808063, 208.7639126876776, 812.9044946001017, 265.40658155815055, 2208.03566170745, 446.3490908361761, 260.63391364642155, 256.1609399551684, 492.894860112155, 9.046498163007648, 1382.3798932062741, 16551.76851963386, 10447.430650796432, 173.2448185714548]

Generation 82: Best Fitness = -3692028666.7495565, Best Individual =

[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.5063774383783, 299.77253657808063, 208.7639126876776, 812.9044946001017, 265.40658155815055, 2208.03566170745, 446.3490908361761, 260.63391364642155, 256.1609399551684, 492.894860112155, 9.046498163007648, 1382.3798932062741, 16551.76851963386, 10447.430650796432, 173.2448185714548]
Generation 83: Best Fitness = -3689524841.8672132, Best Individual =
[4991254306.353332, 63.41236421269042, 2.976321709052205, 836.4879029331403, 834.5660639888896, 448.5848888895305, 299.77253657808063, 211.0491929813066, 1046.6911836999682, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 256.1609399551684, 359.1190537110743, -1.0437374874189018, 1382.3798932062741, 19392.528882023224, 11377.144627159429, 173.2448185714548]
Generation 84: Best Fitness = -3688684017.29699, Best Individual =
[4991254306.353332, 60.21916807948581, 2.976321709052205, 836.4879029331403, 834.5660639888896, 448.5848888895305, 299.77253657808063, 211.0491929813066, 1046.6911836999682, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 256.1609399551684, 497.65532300853573, 9.497327703410015, 1753.4134205301361, 16551.76851963386, 10447.430650796432, 173.2448185714548]
Generation 85: Best Fitness = -3688684017.29699, Best Individual =
[4991254306.353332, 60.21916807948581, 2.976321709052205, 836.4879029331403, 834.5660639888896, 448.5848888895305, 299.77253657808063, 211.0491929813066, 1046.6911836999682, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 256.1609399551684, 497.65532300853573, 9.497327703410015, 1753.4134205301361, 16551.76851963386, 10447.430650796432, 173.2448185714548]
Generation 86: Best Fitness = -3688684017.29699, Best Individual =
[4991254306.353332, 60.21916807948581, 2.976321709052205, 836.4879029331403, 834.5660639888896, 448.5848888895305, 299.77253657808063, 211.0491929813066, 1046.6911836999682, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 256.1609399551684, 497.65532300853573, 9.497327703410015, 1753.4134205301361, 16551.76851963386, 10447.430650796432, 173.2448185714548]
Generation 87: Best Fitness = -3688684017.29699, Best Individual =
[4991254306.353332, 60.21916807948581, 2.976321709052205, 836.4879029331403, 834.5660639888896, 448.5848888895305, 299.77253657808063, 211.0491929813066, 1046.6911836999682, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 256.1609399551684, 497.65532300853573, 9.497327703410015, 1753.4134205301361, 16551.76851963386, 10447.430650796432, 173.2448185714548]
Generation 88: Best Fitness = -3688628899.8400884, Best Individual =
[4991254306.353332, 60.21916807948581, 2.976321709052205, 836.4879029331403, 834.5660639888896, 448.5848888895305, 299.77253657808063, 211.0491929813066, 1046.6911836999682, 381.40713792172403, 2783.418475428156, 438.6132111802663, 423.00305977063056, 256.1609399551684, 497.65532300853573, 9.497327703410015, 1382.3798932062741, 16551.76851963386, 10447.430650796432, 173.2448185714548]
Generation 89: Best Fitness = -3688628899.8400884, Best Individual =
[4991254306.353332, 60.21916807948581, 2.976321709052205, 836.4879029331403, 834.5660639888896, 448.5848888895305, 299.77253657808063, 211.0491929813066, 1046.6911836999682, 381.40713792172403, 2783.418475428156, 438.6132111802663, 423.00305977063056, 256.1609399551684, 497.65532300853573, 9.497327703410015, 1382.3798932062741, 16551.76851963386, 10447.430650796432, 173.2448185714548]
Generation 90: Best Fitness = -3684942538.9030104, Best Individual =

[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.5063774383783, 299.77253657808063, 211.0491929813066, 1046.6911836999682, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 256.1609399551684, 359.1190537110743, -1.0437374874189018, 1484.5411263663625, 19392.528882023224, 11377.144627159429, 173.2448185714548]
Generation 91: Best Fitness = -3684942538.9030104, Best Individual =
[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.5063774383783, 299.77253657808063, 211.0491929813066, 1046.6911836999682, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 256.1609399551684, 359.1190537110743, -1.0437374874189018, 1484.5411263663625, 19392.528882023224, 11377.144627159429, 173.2448185714548]
Generation 92: Best Fitness = -3684942538.9030104, Best Individual =
[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.5063774383783, 299.77253657808063, 211.0491929813066, 1046.6911836999682, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 256.1609399551684, 359.1190537110743, -1.0437374874189018, 1484.5411263663625, 19392.528882023224, 11377.144627159429, 173.2448185714548]
Generation 93: Best Fitness = -3684487346.4686203, Best Individual =
[4991254306.353332, 60.21916807948581, 2.9379569924920172, 836.4879029331403, 834.5660639888896, 449.9396628485834, 299.77253657808063, 211.0491929813066, 1046.6911836999682, 381.40713792172403, 1929.129950377941, 328.7165128679039, 385.76722191678846, 312.1697958413234, 497.65532300853573, 9.497327703410015, 1753.4134205301361, 15328.517967804652, 10447.430650796432, 173.2448185714548]
Generation 94: Best Fitness = -3681805654.893065, Best Individual =
[4991254306.353332, 60.21916807948581, 2.976321709052205, 836.4879029331403, 834.5660639888896, 449.9396628485834, 299.77253657808063, 211.0491929813066, 812.9044946001017, 265.40658155815055, 2596.7315531583235, 438.6132111802663, 304.7213484035832, 385.54039782498165, 497.65532300853573, 9.497327703410015, 1382.3798932062741, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 95: Best Fitness = -3681805654.893065, Best Individual =
[4991254306.353332, 60.21916807948581, 2.976321709052205, 836.4879029331403, 834.5660639888896, 449.9396628485834, 299.77253657808063, 211.0491929813066, 812.9044946001017, 265.40658155815055, 2596.7315531583235, 438.6132111802663, 304.7213484035832, 385.54039782498165, 497.65532300853573, 9.497327703410015, 1382.3798932062741, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 96: Best Fitness = -3681805654.893065, Best Individual =
[4991254306.353332, 60.21916807948581, 2.976321709052205, 836.4879029331403, 834.5660639888896, 449.9396628485834, 299.77253657808063, 211.0491929813066, 812.9044946001017, 265.40658155815055, 2596.7315531583235, 438.6132111802663, 304.7213484035832, 385.54039782498165, 497.65532300853573, 9.497327703410015, 1382.3798932062741, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 97: Best Fitness = -3681258805.5563817, Best Individual =
[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 211.0491929813066, 852.2027243871278, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 256.1609399551684, 359.1190537110743, 9.046498163007648, 1382.3798932062741, 17800.01980843478, 10447.430650796432, 173.2448185714548]
Generation 98: Best Fitness = -3681258805.5563817, Best Individual =

[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 211.0491929813066, 852.2027243871278, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 256.1609399551684, 359.1190537110743, 9.046498163007648, 1382.3798932062741, 17800.01980843478, 10447.430650796432, 173.2448185714548]
Generation 99: Best Fitness = -3680758560.583344, Best Individual =
[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 211.0491929813066, 869.4488819309175, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 293.03648129757426, 497.65532300853573, 9.497327703410015, 1382.3798932062741, 17800.01980843478, 10447.430650796432, 173.2448185714548]
Generation 100: Best Fitness = -3680758560.583344, Best Individual =
[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 211.0491929813066, 869.4488819309175, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 293.03648129757426, 497.65532300853573, 9.497327703410015, 1382.3798932062741, 17800.01980843478, 10447.430650796432, 173.2448185714548]
Generation 101: Best Fitness = -3680758560.583344, Best Individual =
[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 211.0491929813066, 869.4488819309175, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 293.03648129757426, 497.65532300853573, 9.497327703410015, 1382.3798932062741, 17800.01980843478, 10447.430650796432, 173.2448185714548]
Generation 102: Best Fitness = -3680534559.953436, Best Individual =
[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 211.0491929813066, 852.2027243871278, 381.40713792172403, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 256.1609399551684, 491.67048410608936, 9.046498163007648, 1382.3798932062741, 17800.01980843478, 10447.430650796432, 173.2448185714548]
Generation 103: Best Fitness = -3680534559.953436, Best Individual =
[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 211.0491929813066, 852.2027243871278, 381.40713792172403, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 256.1609399551684, 491.67048410608936, 9.046498163007648, 1382.3798932062741, 17800.01980843478, 10447.430650796432, 173.2448185714548]
Generation 104: Best Fitness = -3680534559.953436, Best Individual =
[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 211.0491929813066, 852.2027243871278, 381.40713792172403, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 256.1609399551684, 491.67048410608936, 9.046498163007648, 1382.3798932062741, 17800.01980843478, 10447.430650796432, 173.2448185714548]
Generation 105: Best Fitness = -3680527672.4298444, Best Individual =
[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 211.0491929813066, 852.2027243871278, 265.40658155815055, 2596.7315531583235, 438.6132111802663, 304.7213484035832, 385.54039782498165, 497.65532300853573, 9.497327703410015, 1382.3798932062741, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 106: Best Fitness = -3680527672.4298444, Best Individual =

[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 211.0491929813066, 852.2027243871278, 265.40658155815055, 2596.7315531583235, 438.6132111802663, 304.7213484035832, 385.54039782498165, 497.65532300853573, 9.497327703410015, 1382.3798932062741, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 107: Best Fitness = -3680485669.3340793, Best Individual =
[4991254306.353332, 55.43255678498734, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 211.0491929813066, 852.2027243871278, 274.6101183021991, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 256.1609399551684, 497.65532300853573, 9.497327703410015, 1382.3798932062741, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 108: Best Fitness = -3680485669.3340793, Best Individual =
[4991254306.353332, 55.43255678498734, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 211.0491929813066, 852.2027243871278, 274.6101183021991, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 256.1609399551684, 497.65532300853573, 9.497327703410015, 1382.3798932062741, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 109: Best Fitness = -3679936534.500956, Best Individual =
[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.89066500017975, 211.0491929813066, 852.2027243871278, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 293.03648129757426, 497.65532300853573, 9.497327703410015, 1382.3798932062741, 17800.01980843478, 10447.430650796432, 173.2448185714548]
Generation 110: Best Fitness = -3679936534.500956, Best Individual =
[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.89066500017975, 211.0491929813066, 852.2027243871278, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 293.03648129757426, 497.65532300853573, 9.497327703410015, 1382.3798932062741, 17800.01980843478, 10447.430650796432, 173.2448185714548]
Generation 111: Best Fitness = -3679936534.500956, Best Individual =
[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.89066500017975, 211.0491929813066, 852.2027243871278, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 293.03648129757426, 497.65532300853573, 9.497327703410015, 1382.3798932062741, 17800.01980843478, 10447.430650796432, 173.2448185714548]
Generation 112: Best Fitness = -3679936534.500956, Best Individual =
[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.89066500017975, 211.0491929813066, 852.2027243871278, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 293.03648129757426, 497.65532300853573, 9.497327703410015, 1382.3798932062741, 17800.01980843478, 10447.430650796432, 173.2448185714548]
Generation 113: Best Fitness = -3679936534.500956, Best Individual =
[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.89066500017975, 211.0491929813066, 852.2027243871278, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 293.03648129757426, 497.65532300853573, 9.497327703410015, 1382.3798932062741, 17800.01980843478, 10447.430650796432, 173.2448185714548]
Generation 114: Best Fitness = -3679936534.500956, Best Individual =

[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.89066500017975, 211.0491929813066, 852.2027243871278, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 293.03648129757426, 497.65532300853573, 9.497327703410015, 1382.3798932062741, 17800.01980843478, 10447.430650796432, 173.2448185714548]

Generation 115: Best Fitness = -3679936534.500956, Best Individual =

[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.89066500017975, 211.0491929813066, 852.2027243871278, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 293.03648129757426, 497.65532300853573, 9.497327703410015, 1382.3798932062741, 17800.01980843478, 10447.430650796432, 173.2448185714548]

Generation 116: Best Fitness = -3679936534.500956, Best Individual =

[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.89066500017975, 211.0491929813066, 852.2027243871278, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 293.03648129757426, 497.65532300853573, 9.497327703410015, 1382.3798932062741, 17800.01980843478, 10447.430650796432, 173.2448185714548]

Generation 117: Best Fitness = -3679936534.500956, Best Individual =

[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.89066500017975, 211.0491929813066, 852.2027243871278, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 293.03648129757426, 497.65532300853573, 9.497327703410015, 1382.3798932062741, 17800.01980843478, 10447.430650796432, 173.2448185714548]

Generation 118: Best Fitness = -3679936534.500956, Best Individual =

[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.89066500017975, 211.0491929813066, 852.2027243871278, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 293.03648129757426, 497.65532300853573, 9.497327703410015, 1382.3798932062741, 17800.01980843478, 10447.430650796432, 173.2448185714548]

Generation 119: Best Fitness = -3679936534.500956, Best Individual =

[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.89066500017975, 211.0491929813066, 852.2027243871278, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 293.03648129757426, 497.65532300853573, 9.497327703410015, 1382.3798932062741, 17800.01980843478, 10447.430650796432, 173.2448185714548]

Generation 120: Best Fitness = -3679936534.500956, Best Individual =

[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.89066500017975, 211.0491929813066, 852.2027243871278, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 293.03648129757426, 497.65532300853573, 9.497327703410015, 1382.3798932062741, 17800.01980843478, 10447.430650796432, 173.2448185714548]

Generation 121: Best Fitness = -3679936534.500956, Best Individual =

[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.89066500017975, 211.0491929813066, 852.2027243871278, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 293.03648129757426, 497.65532300853573, 9.497327703410015, 1382.3798932062741, 17800.01980843478, 10447.430650796432, 173.2448185714548]

Generation 122: Best Fitness = -3679936534.500956, Best Individual =

[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.89066500017975, 211.0491929813066, 852.2027243871278, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 293.03648129757426, 497.65532300853573, 9.497327703410015, 1382.3798932062741, 17800.01980843478, 10447.430650796432, 173.2448185714548]

Generation 123: Best Fitness = -3679936534.500956, Best Individual =

[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.89066500017975, 211.0491929813066, 852.2027243871278, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 293.03648129757426, 497.65532300853573, 9.497327703410015, 1382.3798932062741, 17800.01980843478, 10447.430650796432, 173.2448185714548]

Generation 124: Best Fitness = -3679936534.500956, Best Individual =

[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.89066500017975, 211.0491929813066, 852.2027243871278, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 293.03648129757426, 497.65532300853573, 9.497327703410015, 1382.3798932062741, 17800.01980843478, 10447.430650796432, 173.2448185714548]

Generation 125: Best Fitness = -3679936534.500956, Best Individual =

[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.89066500017975, 211.0491929813066, 852.2027243871278, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 293.03648129757426, 497.65532300853573, 9.497327703410015, 1382.3798932062741, 17800.01980843478, 10447.430650796432, 173.2448185714548]

Generation 126: Best Fitness = -3679936534.500956, Best Individual =

[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.89066500017975, 211.0491929813066, 852.2027243871278, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 293.03648129757426, 497.65532300853573, 9.497327703410015, 1382.3798932062741, 17800.01980843478, 10447.430650796432, 173.2448185714548]

Generation 127: Best Fitness = -3679936534.500956, Best Individual =

[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.89066500017975, 211.0491929813066, 852.2027243871278, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 293.03648129757426, 497.65532300853573, 9.497327703410015, 1382.3798932062741, 17800.01980843478, 10447.430650796432, 173.2448185714548]

Generation 128: Best Fitness = -3679936534.500956, Best Individual =

[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.89066500017975, 211.0491929813066, 852.2027243871278, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 293.03648129757426, 497.65532300853573, 9.497327703410015, 1382.3798932062741, 17800.01980843478, 10447.430650796432, 173.2448185714548]

Generation 129: Best Fitness = -3679936534.500956, Best Individual =

[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.89066500017975, 211.0491929813066, 852.2027243871278, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 293.03648129757426, 497.65532300853573, 9.497327703410015, 1382.3798932062741, 17800.01980843478, 10447.430650796432, 173.2448185714548]

Generation 130: Best Fitness = -3679936534.500956, Best Individual =

[4991254306.353332, 63.41236421269042, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.89066500017975, 211.0491929813066, 852.2027243871278, 381.40713792172403, 1515.2872205702045, 438.6132111802663, 385.76722191678846, 293.03648129757426, 497.65532300853573, 9.497327703410015, 1382.3798932062741, 17800.01980843478, 10447.430650796432, 173.2448185714548]

Generation 131: Best Fitness = -3679855238.0780067, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 211.0491929813066, 800.348625862353, 259.9339723293939, 2740.9741772237558, 438.6132111802663, 339.21392867684284, 212.2392667151741, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17622.59182493962, 10447.430650796432, 173.2448185714548]

Generation 132: Best Fitness = -3679855238.0780067, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 211.0491929813066, 800.348625862353, 259.9339723293939, 2740.9741772237558, 438.6132111802663, 339.21392867684284, 212.2392667151741, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17622.59182493962, 10447.430650796432, 173.2448185714548]

Generation 133: Best Fitness = -3679855238.0780067, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 211.0491929813066, 800.348625862353, 259.9339723293939, 2740.9741772237558, 438.6132111802663, 339.21392867684284, 212.2392667151741, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17622.59182493962, 10447.430650796432, 173.2448185714548]

Generation 134: Best Fitness = -3679855238.0780067, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 211.0491929813066, 800.348625862353, 259.9339723293939, 2740.9741772237558, 438.6132111802663, 339.21392867684284, 212.2392667151741, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17622.59182493962, 10447.430650796432, 173.2448185714548]

Generation 135: Best Fitness = -3679855238.0780067, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 211.0491929813066, 800.348625862353, 259.9339723293939, 2740.9741772237558, 438.6132111802663, 339.21392867684284, 212.2392667151741, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17622.59182493962, 10447.430650796432, 173.2448185714548]

Generation 136: Best Fitness = -3679855238.0780067, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 211.0491929813066, 800.348625862353, 259.9339723293939, 2740.9741772237558, 438.6132111802663, 339.21392867684284, 212.2392667151741, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17622.59182493962, 10447.430650796432, 173.2448185714548]

Generation 137: Best Fitness = -3679855238.0780067, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 211.0491929813066, 800.348625862353, 259.9339723293939, 2740.9741772237558, 438.6132111802663, 339.21392867684284, 212.2392667151741, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17622.59182493962, 10447.430650796432, 173.2448185714548]

Generation 138: Best Fitness = -3679855238.0780067, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 211.0491929813066, 800.348625862353, 259.9339723293939, 2740.9741772237558, 438.6132111802663, 339.21392867684284, 212.2392667151741, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 139: Best Fitness = -3679855238.0780067, Best Individual =
[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 211.0491929813066, 800.348625862353, 259.9339723293939, 2740.9741772237558, 438.6132111802663, 339.21392867684284, 212.2392667151741, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 140: Best Fitness = -3679855238.0780067, Best Individual =
[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 211.0491929813066, 800.348625862353, 259.9339723293939, 2740.9741772237558, 438.6132111802663, 339.21392867684284, 212.2392667151741, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 141: Best Fitness = -3679855238.0780067, Best Individual =
[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 211.0491929813066, 800.348625862353, 259.9339723293939, 2740.9741772237558, 438.6132111802663, 339.21392867684284, 212.2392667151741, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 142: Best Fitness = -3679855238.0780067, Best Individual =
[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 211.0491929813066, 800.348625862353, 259.9339723293939, 2740.9741772237558, 438.6132111802663, 339.21392867684284, 212.2392667151741, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 143: Best Fitness = -3679855238.0780067, Best Individual =
[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 211.0491929813066, 800.348625862353, 259.9339723293939, 2740.9741772237558, 438.6132111802663, 339.21392867684284, 212.2392667151741, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 144: Best Fitness = -3679855238.0780067, Best Individual =
[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 211.0491929813066, 800.348625862353, 259.9339723293939, 2740.9741772237558, 438.6132111802663, 339.21392867684284, 212.2392667151741, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 145: Best Fitness = -3672929754.7770095, Best Individual =
[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]
Generation 146: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 147: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 148: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 149: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 150: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 151: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 152: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 153: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 154: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 155: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 156: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 157: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 158: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 159: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 160: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 161: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 162: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 163: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 164: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 165: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 166: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 167: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 168: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 169: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 170: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 171: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 172: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 173: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 174: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 175: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 176: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 177: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 178: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]
 Generation 179: Best Fitness = -3672929754.7770095, Best Individual =
 [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]
 Generation 180: Best Fitness = -3672929754.7770095, Best Individual =
 [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]
 Generation 181: Best Fitness = -3672929754.7770095, Best Individual =
 [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]
 Generation 182: Best Fitness = -3672929754.7770095, Best Individual =
 [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]
 Generation 183: Best Fitness = -3672929754.7770095, Best Individual =
 [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]
 Generation 184: Best Fitness = -3672929754.7770095, Best Individual =
 [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]
 Generation 185: Best Fitness = -3672929754.7770095, Best Individual =
 [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]
 Generation 186: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 187: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 188: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 189: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 190: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 191: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 192: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 193: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 194: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 195: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 196: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 197: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 198: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 199: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 200: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 201: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 202: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 203: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 204: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 205: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 206: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 207: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 208: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 209: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 210: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 211: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 212: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 213: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 214: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 215: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 216: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 217: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 218: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 219: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 220: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 221: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 222: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 223: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 224: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 225: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 226: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]
Generation 227: Best Fitness = -3672929754.7770095, Best Individual =
[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]
Generation 228: Best Fitness = -3672929754.7770095, Best Individual =
[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]
Generation 229: Best Fitness = -3672929754.7770095, Best Individual =
[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]
Generation 230: Best Fitness = -3672929754.7770095, Best Individual =
[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]
Generation 231: Best Fitness = -3672929754.7770095, Best Individual =
[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]
Generation 232: Best Fitness = -3672929754.7770095, Best Individual =
[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]
Generation 233: Best Fitness = -3672929754.7770095, Best Individual =
[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]
Generation 234: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 235: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 236: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 237: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 238: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 239: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 240: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 241: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 242: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 243: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 244: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 245: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 246: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 247: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 248: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 249: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 250: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 251: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 252: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 253: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 254: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 255: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 256: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 257: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 258: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]
Generation 259: Best Fitness = -3672929754.7770095, Best Individual =
[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]
Generation 260: Best Fitness = -3672929754.7770095, Best Individual =
[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]
Generation 261: Best Fitness = -3672929754.7770095, Best Individual =
[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]
Generation 262: Best Fitness = -3672929754.7770095, Best Individual =
[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]
Generation 263: Best Fitness = -3672929754.7770095, Best Individual =
[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]
Generation 264: Best Fitness = -3672929754.7770095, Best Individual =
[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]
Generation 265: Best Fitness = -3672929754.7770095, Best Individual =
[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]
Generation 266: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]
Generation 267: Best Fitness = -3672929754.7770095, Best Individual =
[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]
Generation 268: Best Fitness = -3672929754.7770095, Best Individual =
[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]
Generation 269: Best Fitness = -3672929754.7770095, Best Individual =
[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]
Generation 270: Best Fitness = -3672929754.7770095, Best Individual =
[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]
Generation 271: Best Fitness = -3672929754.7770095, Best Individual =
[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]
Generation 272: Best Fitness = -3672929754.7770095, Best Individual =
[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]
Generation 273: Best Fitness = -3672929754.7770095, Best Individual =
[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]
Generation 274: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 275: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 276: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 277: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 278: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 279: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 280: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 281: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 282: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 283: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 284: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 285: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 286: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 287: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 288: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 289: Best Fitness = -3672929754.7770095, Best Individual = [4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]

Generation 290: Best Fitness = -3672929754.7770095, Best Individual =

[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]
Generation 291: Best Fitness = -3672929754.7770095, Best Individual =
[4991254306.353332, 57.68651311939331, 2.976321709052205, 835.1492890941504, 834.5660639888896, 449.9396628485834, 299.77253657808063, 212.56234571167028, 808.5967082378584, 329.41624052584507, 2740.9741772237558, 438.6132111802663, 385.76722191678846, 239.41676422230225, 497.65532300853573, 9.993415687972195, 1382.3798932062741, 17569.385445898297, 10447.954177770627, 173.2448185714548]
Generation 292: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 293: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 294: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 295: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 296: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 297: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 298: Best Fitness = -3672925586.397032, Best Individual =

[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 299: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 300: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 301: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 302: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 303: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 304: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 305: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 306: Best Fitness = -3672925586.397032, Best Individual =

[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 307: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 308: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 309: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 310: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 311: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 312: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 313: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 314: Best Fitness = -3672925586.397032, Best Individual =

[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 315: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 316: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 317: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 318: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 319: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 320: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 321: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 322: Best Fitness = -3672925586.397032, Best Individual =

[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 323: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 324: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 325: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 326: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 327: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 328: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 329: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 330: Best Fitness = -3672925586.397032, Best Individual =

[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 331: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 332: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 333: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 334: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 335: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 336: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 337: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 338: Best Fitness = -3672925586.397032, Best Individual =

[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 339: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 340: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 341: Best Fitness = -3672925586.397032, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 836.0156669591574, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.34826615705722, 800.348625862353, 251.69691592129155, 2740.9741772237558, 423.68322398437505, 387.6040746283569, 236.09944888627754, 499.7448946930915, 9.993415687972195, 1359.9703613288384, 17622.59182493962, 10447.430650796432, 173.2448185714548]
Generation 342: Best Fitness = -3670889013.2444167, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 343: Best Fitness = -3670889013.2444167, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 344: Best Fitness = -3670889013.2444167, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 345: Best Fitness = -3670889013.2444167, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 346: Best Fitness = -3670889013.2444167, Best Individual =

[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 347: Best Fitness = -3670889013.2444167, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 348: Best Fitness = -3670889013.2444167, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 349: Best Fitness = -3670889013.2444167, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 350: Best Fitness = -3670889013.2444167, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 351: Best Fitness = -3670889013.2444167, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 352: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 353: Best Fitness = -3670889013.2444167, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 354: Best Fitness = -3670889013.2444167, Best Individual =

[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 355: Best Fitness = -3670889013.2444167, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 356: Best Fitness = -3670889013.2444167, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 357: Best Fitness = -3670889013.2444167, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 358: Best Fitness = -3670889013.2444167, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 359: Best Fitness = -3670889013.2444167, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 360: Best Fitness = -3670889013.2444167, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 361: Best Fitness = -3670889013.2444167, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 362: Best Fitness = -3670889013.2444167, Best Individual =

[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
 Generation 363: Best Fitness = -3670889013.2444167, Best Individual =
 [4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
 Generation 364: Best Fitness = -3670889013.2444167, Best Individual =
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 Generation 365: Best Fitness = -3670889013.2444167, Best Individual =
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 Generation 366: Best Fitness = -3670889013.2444167, Best Individual =
 [4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
 Generation 367: Best Fitness = -3670889013.2444167, Best Individual =
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 Generation 368: Best Fitness = -3670889013.2444167, Best Individual =
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 Generation 369: Best Fitness = -3670889013.2444167, Best Individual =
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 Generation 370: Best Fitness = -3670889013.2444167, Best Individual =

[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 371: Best Fitness = -3670889013.2444167, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 372: Best Fitness = -3670889013.2444167, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 373: Best Fitness = -3670889013.2444167, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 374: Best Fitness = -3670889013.2444167, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 375: Best Fitness = -3670889013.2444167, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 376: Best Fitness = -3670889013.2444167, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 377: Best Fitness = -3670889013.2444167, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 378: Best Fitness = -3670889013.2444167, Best Individual =

[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 379: Best Fitness = -3670889013.2444167, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 380: Best Fitness = -3670889013.2444167, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 381: Best Fitness = -3670889013.2444167, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 382: Best Fitness = -3670889013.2444167, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 383: Best Fitness = -3670889013.2444167, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 384: Best Fitness = -3670889013.2444167, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 385: Best Fitness = -3670889013.2444167, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 386: Best Fitness = -3670889013.2444167, Best Individual =

[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 387: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 388: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 389: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 390: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 391: Best Fitness = -3670889013.2444167, Best Individual =
[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 392: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 393: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 394: Best Fitness = -3670889013.2444167, Best Individual =

[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
 Generation 395: Best Fitness = -3670889013.2444167, Best Individual =
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 Generation 396: Best Fitness = -3670889013.2444167, Best Individual =
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 Generation 397: Best Fitness = -3670889013.2444167, Best Individual =
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 Generation 398: Best Fitness = -3670889013.2444167, Best Individual =
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 Generation 399: Best Fitness = -3670889013.2444167, Best Individual =
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 Generation 400: Best Fitness = -3670889013.2444167, Best Individual =
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 Generation 401: Best Fitness = -3670889013.2444167, Best Individual =
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 Generation 402: Best Fitness = -3670889013.2444167, Best Individual =

[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 403: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 404: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 405: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 406: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 407: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 408: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 409: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 410: Best Fitness = -3670889013.2444167, Best Individual =

[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 411: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 412: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 413: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 414: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 415: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 416: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 417: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 418: Best Fitness = -3670889013.2444167, Best Individual =

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Generation 420: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 421: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 422: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 423: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 424: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 425: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 426: Best Fitness = -3670889013.2444167, Best Individual =

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Generation 427: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 428: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 429: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 430: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 431: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 432: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 433: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 434: Best Fitness = -3670889013.2444167, Best Individual =

[4991254306.353332, 57.68651311939331, 2.8738576126879196, 834.8279673454947, 834.5660639888896, 449.9396628485834, 299.89066500017975, 212.54931057185462, 800.348625862353, 250.7114309274385, 2740.9741772237558, 423.68322398437505, 446.5330066280377, 236.09944888627754, 499.7448946930915, 9.981207272092064, 1359.9703613288384, 17085.840707679006, 10447.430650796432, 173.2448185714548]
Generation 435: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 436: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 437: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 438: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 439: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 440: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 441: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 443: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 444: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 445: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 446: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 447: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 448: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 449: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 450: Best Fitness = -3670889013.2444167, Best Individual =

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Generation 451: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 452: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 453: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 454: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 455: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 456: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 457: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 458: Best Fitness = -3670889013.2444167, Best Individual =

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Generation 459: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 460: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 461: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 462: Best Fitness = -3670889013.2444167, Best Individual =
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Generation 496: Best Fitness = -3670841142.8216667, Best Individual =
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X17 =19815.3206424328
X18 =10407.9750676304
```

```
model9 = joblib.load('/home/houssam/internship project/trained_models/ Ammonia_
↳ Comp Mass Flow_Pipeline.pkl')

mass = predict_Mass(X1,X2,X10,X11,X12,X13,X16,X17,X18,model1,model9)
print(mass)
```

```
171.81617710299793
```

```
[ ]: #save the best observations
```

```
[ ]: sorted_population = [ind for ind, fit in sorted(zip(population, fitnesses),
↳ key=lambda x: x[1], reverse=True)]
elites = sorted_population[:50]
elites_fitnesses = [fitness_function(ind) for ind in elites]
elites_ammonia_mass =
↳ [predict_Mass(individual[1],individual[2],individual[10],individual[11],individual[12],indi
↳ for individual in elites]
output_data = {'NG FEED - Comp Mass Flow (KJ/h)':[elite[1] for elite in elites],
'Ratio (S/C)':[elite[2] for elite in elites],
'Reformer inlet temperature (°C)':[elite[3] for elite in elites],
'Reformer outlet temperature (°C)':[elite[4] for elite in
↳ elites],
'HTS - Inlet temperature (°C)':[elite[5] for elite in elites],
'LTS - Inlet temperature (°C)':[elite[6] for elite in elites],
```

```

        'Condensor inlet temperature (°C)':[elite[7] for elite in elites],
        'Condensor inlet pressure (kPa)':[elite[8] for elite in elites],
        'Methanator inlet temperature (°C)':[elite[9] for elite in elites],
        'LTS - Inlet pressure (kPa)':[elite[10] for elite in elites],
        'Hydrogen inlet temperature (°C)':[elite[11] for elite in elites],
        'Nitrogen inlet temperature (°C)':[elite[12] for elite in elites],
        'K-100 - Delta P (kPa)':[elite[13] for elite in elites],
        'Synthesis gas inlet temperature (°C)':[elite[14] for elite in elites],
        'Separator inlet temperature (°C)':[elite[15] for elite in elites],
        'K-102 - Delta P (kPa)':[elite[16] for elite in elites],
        'Synthesis gas inlet pressure (kPa)':[elite[17] for elite in elites],
        'Separator inlet pressure (kPa)':[elite[18] for elite in elites],
        'total_energy':[-value for value in elites_fitnesses],
        'Ammonia_mass':elites_ammonia_mass}
output_data = pd.DataFrame(output_data)
output_data.sort_values(by='total_energy', ascending=True,inplace = True)
output_data.head(11)
output_data.to_excel('output2.xlsx', index=False)

```

```
[ ]: #genetic algo with much flexible ranges
```

```

[1]: #define the population size and number of generations
POPULATION_SIZE = 100
NUM_GENERATIONS = 500
MUTATION_RATE = 0.5
ELITISM_SIZE = 20 # Number of top individuals to carry over

# Define parameter ranges
ranges_dict = {col: {'Min': df[col].min(), 'Max': df[col].max()}} for col in df.
columns}

# It will helps us to go from index to name
predictions_list =['total_energy','NG FEED - Comp Mass Flow (KJ/h)','Ratio (S/
°C)','Reformer inlet temperature (°C)',
    'Reformer outlet temperature (°C)','HTS - Inlet temperature (°C)','LTS -
Inlet temperature (°C)','Condensor inlet temperature (°C)',

```



```

    'Condensor inlet pressure (kPa)', 'Methanator inlet temperature (°C)', 'LTS -
↳ Inlet pressure (kPa)',
    'Hydrogen inlet temperature (°C)', 'Nitrogen inlet temperature (°C)', 'K-100
↳ - Delta P (kPa)', 'Synthesis gas inlet temperature (°C)', 'Separator inlet
↳ temperature (°C)',
    'K-102 - Delta P (kPa)', 'Synthesis gas inlet pressure (kPa)', 'Separator
↳ inlet pressure (kPa)'
    , 'Ammonia - Comp Mass Flow (kg/h)']

#load the data
#S_models
modelS2 = joblib.load('/home/houssam/internship project/trained_models/S-2
↳ stream temperature_Pipeline.pkl')
modelS3 = joblib.load('/home/houssam/internship project/trained_models/S-3
↳ stream temperature_Pipeline.pkl')

#energy
model1 = joblib.load('/home/houssam/internship project/trained_models/
↳ Mixed_Steam_Pipeline.pkl')
model2 = joblib.load('/home/houssam/internship project/trained_models/
↳ Reformer_Energy_Pipeline.pkl')
model3 = joblib.load('/home/houssam/internship project/trained_models/Q-102 -
↳ Energy_Pipeline.pkl')
model4 = joblib.load('/home/houssam/internship project/trained_models/E-1-
↳ Energy_Pipeline.pkl')
model5 = joblib.load('/home/houssam/internship project/trained_models/E-2-
↳ Energy_Pipeline.pkl')
model6 = joblib.load('/home/houssam/internship project/trained_models/E-3-
↳ Energy_Pipeline.pkl')
model7 = joblib.load('/home/houssam/internship project/trained_models/E-4-
↳ Energy_Pipeline.pkl')
model8 = joblib.load('/home/houssam/internship project/trained_models/
↳ E-separator- Energy_Pipeline.pkl')

#Ammonia Mass
model9 = joblib.load('/home/houssam/internship project/trained_models/ Ammonia
↳ - Comp Mass Flow_Pipeline.pkl')

#CO2 emission
model10 = joblib.load('/home/houssam/internship project/trained_models/CO2
↳ emissions-Comp Mass Flow_Pipeline.pkl')

# Load the trained model
'''
total_energy = X0

```

```

NG FEED - Comp Mass Flow (KJ/h) = X1
Ratio (S/C) = X2
Reformer inlet temperature (°C) = X3
Reformer outlet temperature (°C) = X4
HTS - Inlet temperature (°C) = X5
LTS - Inlet temperature (°C) = X6
Condensor inlet temperature (°C) = X7
Condensor inlet pressure (kPa) = X8
Methanator inlet temperature (°C) = X9

Mixed Steam- Comp Mass Flow (KJ/h) = model1.predict(X1,X2)
LTS - Inlet pressure (kPa) = X10
Hydrogen inlet temperature (°C) = X11
Nitrogen inlet temperature (°C) = X12
K-100 - Delta P (kPa) = X13
S-2 stream temperature (°C) = modelS2.predict(X11,X12,X13)
S-3 stream temperature (°C) = modelS3.predict(X1,X2,X10,X11,X12,X13)
Synthesis gas inlet temperature (°C) = X14
Separator inlet temperature (°C) = X15
K-102 - Delta P (kPa) = X16
Synthesis gas inlet pressure (kPa) = X17
Separator inlet pressure (kPa) = X18

'Mixed Steam- Comp Mass Flow (KJ/h)', 'LTS - Inlet pressure (kPa)', 'S-2 stream_
↳temperature (°C)', 'S-3 stream temperature (°C)',
    'K-100 - Delta P (kPa)', 'K-102 - Delta P (kPa)', 'Synthesis gas inlet_
↳pressure (kPa)', 'Separator inlet pressure (kPa)'

'''
def_
↳predict_energy(X1,X2,X3,X4,X5,X6,X7,X8,X9,X10,X11,X12,X13,X14,X15,model1,model2,model3,mode
↳
    Total_energy = 0

    Mixed_Steam = model1.predict([[X1,X2]])[0]
    S2 = modelS2.predict([[X11,X12,X13]])[0]
    S3 = modelS3.predict([[X11,X12,X13,Mixed_Steam,X10]])[0]
    Total_energy += model2.predict([[X3,X4]])[0]
    Total_energy += 4227654.693147199*X3 - 1507124898.3797724
    Total_energy += model3.predict([[X4,X5]])[0]
    Total_energy += -3082293.5646714796*X5 + 1077098983.4823322
    Total_energy += -2640932.047069013*X6 + 953513207.3016243
    Total_energy += -2640920.0637122733*X6 + 559925580.9712701
    Total_energy += -2318202.284356667*X7 + 509146085.55179334
    Total_energy += -2318641.333250072*X7 + 20190755.60255745
    Total_energy += 11485.991953742923*X8 - 3851589.240937948
    Total_energy += 2750563.303329907*X9 - 107470876.00437343

```

```

Total_energy += -2750561.059656047*X9 + 770251086.4382389
Total_energy += model4.predict([[Mixed_Steam,X10,X11,X13,S2]])[0]
Total_energy += model5.predict([[S2,S3,X13]])[0]
Total_energy += model6.predict([[Mixed_Steam,X10,X14]])[0]
Total_energy += -2579.93651444447*X14 + 747019.723192719
Total_energy += model7.predict([[Mixed_Steam, X10,X15]])[0]
Total_energy += model8.predict([[X15]])[0]

return Total_energy

def predict_Mass(X1,X2,X10,X11,X12,X13,X16,X17,X18,model1,model9):
    Mixed_Steam = model1.predict([[X1,X2]])[0]
    S2 = modelS2.predict([[X11,X12,X13]])[0]
    S3 = modelS3.predict([[X11,X12,X13,Mixed_Steam,X10]])[0]
    output_mass = model9.predict([[Mixed_Steam,X10,S2,S3,X13,X16,X17,X18]])[0]
    return output_mass

def CO2_emission(X1,X2,X10,model1,model10):
    Mixed_Steam = model1.predict([[X1,X2]])[0]
    output = model10.predict([[Mixed_Steam,X10]])[0]
    return output

def create_population(df,size,predictures_list):
    sorted_df = df.sort_values(by='total_energy').head(size)
    sorted_df = sorted_df[predictures_list]
    return sorted_df.values.tolist()

def fitness_function(individual):
    predicted_mass = □
    ↪predict_Mass(individual[1],individual[2],individual[10],individual[11],individual[12],indiv
    predicted_CO2_mass = □
    ↪CO2_emission(individual[1],individual[2],individual[10],model1,model10)
    if predicted_mass > 164 and 172 > predicted_mass and predicted_CO2_mass□
    ↪ 256:
        predicted_energy = □
        ↪predict_energy(individual[1],individual[2],individual[3],individual[4],individual[5],indivi
        return -predicted_energy
    else:
        return -float('inf')

def tournament_selection(population, fitnesses, k=3):
    selected = random.sample(list(zip(population, fitnesses)), k)

```

```

    return min(selected, key=lambda item: item[1])[0]

def crossover(parent1, parent2):
    crossover_point = random.randint(2, len(parent1) - 2)
    child1 = parent1[:crossover_point] + parent2[crossover_point:]
    child2 = parent2[:crossover_point] + parent1[crossover_point:]
    return child1, child2

def mutate(individual):
    """Mutate an individual's parameters."""
    if random.random() < MUTATION_RATE:
        mutation_index = random.randint(1, len(individual) - 2)
        Max = ranges_dict[predictures_list[mutation_index]]['Max']
        Min = ranges_dict[predictures_list[mutation_index]]['Min']
        individual[mutation_index] = random.uniform(Min,Max)
    return individual

#the main algo

# Create initial population (type: df)
population = create_population(df,POPULATION_SIZE,predictures_list)
fitnesses = [-ind[0] for ind in population]
# Sort the population by fitness and preserve the top individuals
sorted_population = [ind for ind, fit in sorted(zip(population, fitnesses),
    ↪key=lambda x: x[1], reverse=True)]
elites = sorted_population[:ELITISM_SIZE]
elites_fitnesses = [fitness_function(ind) for ind in elites]

# Create a new population through selection, crossover, and mutation
new_population = elites # Start with the best individuals

while len(new_population) < POPULATION_SIZE:
    parent1 = tournament_selection(elites, elites_fitnesses)
    parent2 = tournament_selection(elites, elites_fitnesses)
    child1, child2 = crossover(parent1, parent2)
    new_population.append(mutate(child1))
    if len(new_population) < POPULATION_SIZE:
        new_population.append(mutate(child2))

```

```

# Replace the old population with the new population
population = new_population

# Print the best individual of the current generation
best_individual = max(population, key=fitness_function)
best_fitness = fitness_function(best_individual)
print(f'Generation {0}: Best Fitness = {best_fitness}, Best Individual = {best_individual}')

for generation in range(1, NUM_GENERATIONS):
    # Evaluate fitness of each individual
    fitnesses = [fitness_function(ind) for ind in population]

    # Sort the population by fitness and preserve the top individuals
    sorted_population = [ind for ind, fit in sorted(zip(population, fitnesses),
key=lambda x: x[1], reverse=True)]
    elites = sorted_population[:ELITISM_SIZE]
    elites_fitnesses = [fitness_function(ind) for ind in elites]

    # Create a new population through selection, crossover, and mutation
    new_population = elites # Start with the best individuals

    while len(new_population) < POPULATION_SIZE:
        parent1 = tournament_selection(elites, elites_fitnesses)
        parent2 = tournament_selection(elites, elites_fitnesses)
        child1, child2 = crossover(parent1, parent2)
        new_population.append(mutate(child1))
        if len(new_population) < POPULATION_SIZE:
            new_population.append(mutate(child2))

    # Replace the old population with the new population
    population = new_population

    # Print the best individual of the current generation
    best_individual = max(population, key=fitness_function)
    best_fitness = fitness_function(best_individual)
    print(f'Generation {generation}: Best Fitness = {best_fitness}, Best
Individual = {best_individual}')

```

```

-----
NameError                                Traceback (most recent call last)
Cell In[1], line 9
      5 ELITISM_SIZE = 20 # Number of top individuals to carry over
      8 # Define parameter ranges
----> 9 ranges_dict = {col: {'Min': df[col].min(), 'Max': df[col].max()} for col
    ↪in df.columns}
     10 ranges_dict['NG FEED - Comp Mass Flow (KJ/h)']={'Min':50,'Max':100}
     11 ranges_dict['Ratio (S/C)']={'Min':2.5,'Max':5}

NameError: name 'df' is not defined

```

```

[8]: sorted_population = [ind for ind, fit in sorted(zip(population, fitnesses),
    ↪key=lambda x: x[1], reverse=True)]
elites = sorted_population[:50]
elites_fitnesses = [fitness_function(ind) for ind in elites]
elites_ammonia_mass =
    ↪[predict_Mass(individual[1],individual[2],individual[10],individual[11],individual[12],indi
    ↪for individual in elites]
output_data = {'NG FEED - Comp Mass Flow (KJ/h)':[elite[1] for elite in elites],
    'Ratio (S/C)':[elite[2] for elite in elites],
    'Reformer inlet temperature (°C)':[elite[3] for elite in elites],
    'Reformer outlet temperature (°C)':[elite[4] for elite in
    ↪elites],
    'HTS - Inlet temperature (°C)':[elite[5] for elite in elites],
    'LTS - Inlet temperature (°C)':[elite[6] for elite in elites],
    'Condensor inlet temperature (°C)':[elite[7] for elite in
    ↪elites],
    'Condensor inlet pressure (kPa)':[elite[8] for elite in elites],
    'Methanator inlet temperature (°C)':[elite[9] for elite in
    ↪elites],
    'LTS - Inlet pressure (kPa)':[elite[10] for elite in elites],
    'Hydrogen inlet temperature (°C)':[elite[11] for elite in
    ↪elites],
    'Nitrogen inlet temperature (°C)':[elite[12] for elite in
    ↪elites],
    'K-100 - Delta P (kPa)':[elite[13] for elite in elites],
    'Synthesis gas inlet temperature (°C)':[elite[14] for elite in
    ↪elites],
    'Separator inlet temperature (°C)':[elite[15] for elite in
    ↪elites],
    'K-102 - Delta P (kPa)':[elite[16] for elite in elites],
    'Synthesis gas inlet pressure (kPa)':[elite[17] for elite in
    ↪elites],
    'Separator inlet pressure (kPa)':[elite[18] for elite in elites],

```

```

        'total_energy':[-value for value in elites_fitnesses],
        'Ammonia_mass':elites_ammonia_mass}
output_data = pd.DataFrame(output_data)
output_data.sort_values(by='total_energy', ascending=True,inplace = True)
output_data.head(11)

```

```

[8]:      NG FEED - Comp Mass Flow (KJ/h)  Ratio (S/C)  \
0                65.096384      2.523044
21               64.168356      2.503846
23               64.168356      2.503846
27               64.168356      2.503846
28               64.168356      2.503846
31               64.168356      2.503846
34               64.168356      2.503846
35               64.168356      2.503846
20               64.168356      2.503846
37               64.168356      2.503846
40               64.168356      2.503846

      Reformer inlet temperature (°C)  Reformer outlet temperature (°C)  \
0                754.060969                800.068371
21               751.228014                800.076341
23               751.228014                800.076341
27               751.228014                800.076341
28               751.228014                800.076341
31               751.228014                800.076341
34               751.228014                800.076341
35               751.228014                800.076341
20               751.228014                800.076341
37               751.228014                800.076341
40               751.228014                800.076341

      HTS - Inlet temperature (°C)  LTS - Inlet temperature (°C)  \
0                449.975203                299.946937
21               449.975203                299.946937
23               449.975203                299.946937
27               449.975203                299.946937
28               449.975203                299.946937
31               449.975203                299.946937
34               449.975203                299.946937
35               449.975203                299.946937
20               449.975203                299.946937
37               449.975203                299.946937
40               449.975203                299.946937

      Condensor inlet temperature (°C)  Condensor inlet pressure (kPa)  \
0                249.738881                801.03633

```

21	249.738881	801.03633
23	249.738881	801.03633
27	249.738881	801.03633
28	249.738881	801.03633
31	249.738881	801.03633
34	249.738881	801.03633
35	249.738881	801.03633
20	249.738881	801.03633
37	249.738881	801.03633
40	249.738881	801.03633

	Methanator inlet temperature (°C)	LTS - Inlet pressure (kPa) \
0	338.741795	2349.347471
21	250.609777	2720.205457
23	250.609777	2742.860490
27	250.609777	2720.205457
28	250.609777	2555.886957
31	250.609777	2720.205457
34	250.609777	2541.427029
35	250.609777	2742.860490
20	250.609777	2742.860490
37	250.609777	2720.205457
40	250.609777	2555.886957

	Hydrogen inlet temperature (°C)	Nitrogen inlet temperature (°C) \
0	392.143019	383.207987
21	392.143019	379.229305
23	392.143019	379.229305
27	392.143019	421.339522
28	392.143019	379.229305
31	392.143019	379.229305
34	392.143019	379.229305
35	392.143019	379.229305
20	392.143019	379.229305
37	392.143019	379.229305
40	392.143019	379.229305

	K-100 - Delta P (kPa)	Synthesis gas inlet temperature (°C) \
0	229.427339	499.662779
21	232.920594	499.662779
23	232.920594	499.662779
27	232.920594	499.662779
28	232.920594	499.662779
31	232.920594	499.662779
34	232.920594	499.662779
35	232.920594	499.662779
20	232.920594	499.662779

37	232.920594	499.662779
40	232.920594	499.662779

	Separator inlet temperature (°C)	K-102 - Delta P (kPa) \
0	9.955674	1742.325346
21	9.974162	1742.325346
23	9.974162	1742.325346
27	9.974162	1330.944755
28	9.974162	1742.325346
31	9.974162	1330.944755
34	9.974162	1742.325346
35	9.974162	1742.325346
20	9.974162	1742.325346
37	9.974162	1742.325346
40	9.974162	1853.617287

	Synthesis gas inlet pressure (kPa)	Separator inlet pressure (kPa) \
0	19815.320642	10407.975068
21	17141.303603	10407.975068
23	17141.303603	10407.975068
27	17141.303603	10407.975068
28	17141.303603	10407.975068
31	17141.303603	10407.975068
34	17141.303603	10407.975068
35	17141.303603	10407.975068
20	17141.303603	10407.975068
37	16891.129218	10407.975068
40	17141.303603	10266.039070

	total_energy	Ammonia_mass
0	2.850249e+09	171.873768
21	2.850312e+09	171.998852
23	2.850312e+09	171.998852
27	2.850312e+09	171.558621
28	2.850312e+09	171.998852
31	2.850312e+09	171.558621
34	2.850312e+09	171.998852
35	2.850312e+09	171.998852
20	2.850312e+09	171.998852
37	2.850312e+09	171.998852
40	2.850312e+09	171.986879

```
[9]: output_data.to_excel('output.xlsx', index=False)
```

Generation 58: Best Fitness = -3561533443.0976515, Best Individual = [4816216819.912155, 67.6119, 2.1726849623877857, 861.422, 832.0011, 442.8251690213361, 263.73857366736877, 211.4965920436418, 1533.6201, 322.2459720353735, 2042.7291, 229.1899142129273,

304.398, 459.681545973852, 545.398, 108.20743417124231, 473.5377, 551.9091293752036,
28.305766282883912, 1859.4227, 2332.9604, 20536.5407386704, 10536.5407386704,
167.659842449638]